

# **Large increases in public R&D investment are needed to avoid declines of US agricultural productivity**

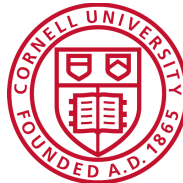
**Ariel Ortiz-Bobea**

Associate Professor of Applied Economics and Policy

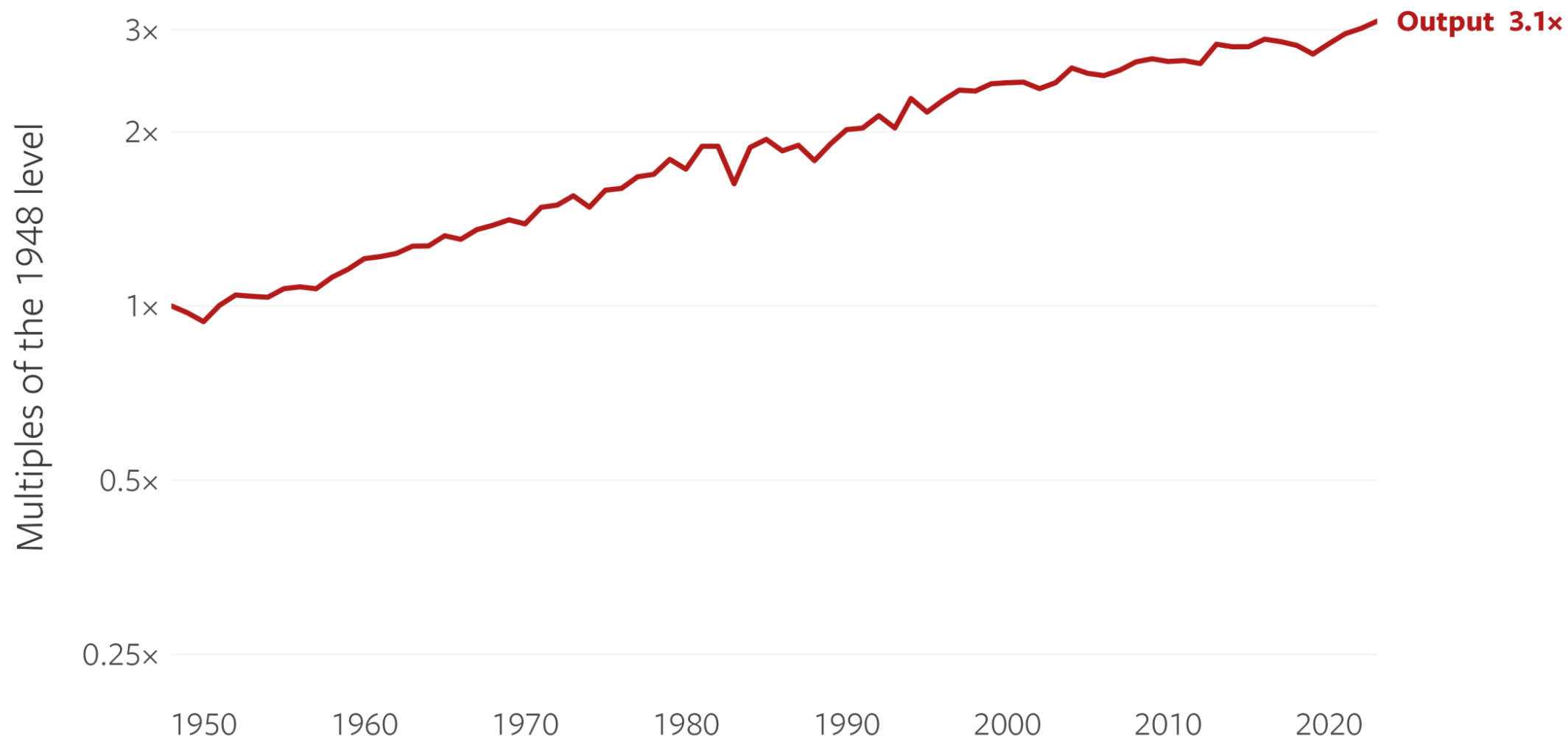
Dyson School of Applied Economics and Management · Brooks School of Public Policy · Cornell University

Talk based on a 2025 PNAS article co-authored with  
Robert Chambers (Maryland), Yurou He (Cornell) and David Lobell (Stanford)

agInnovation Fall Meeting · Grand Junction, Colorado · September 22, 2026

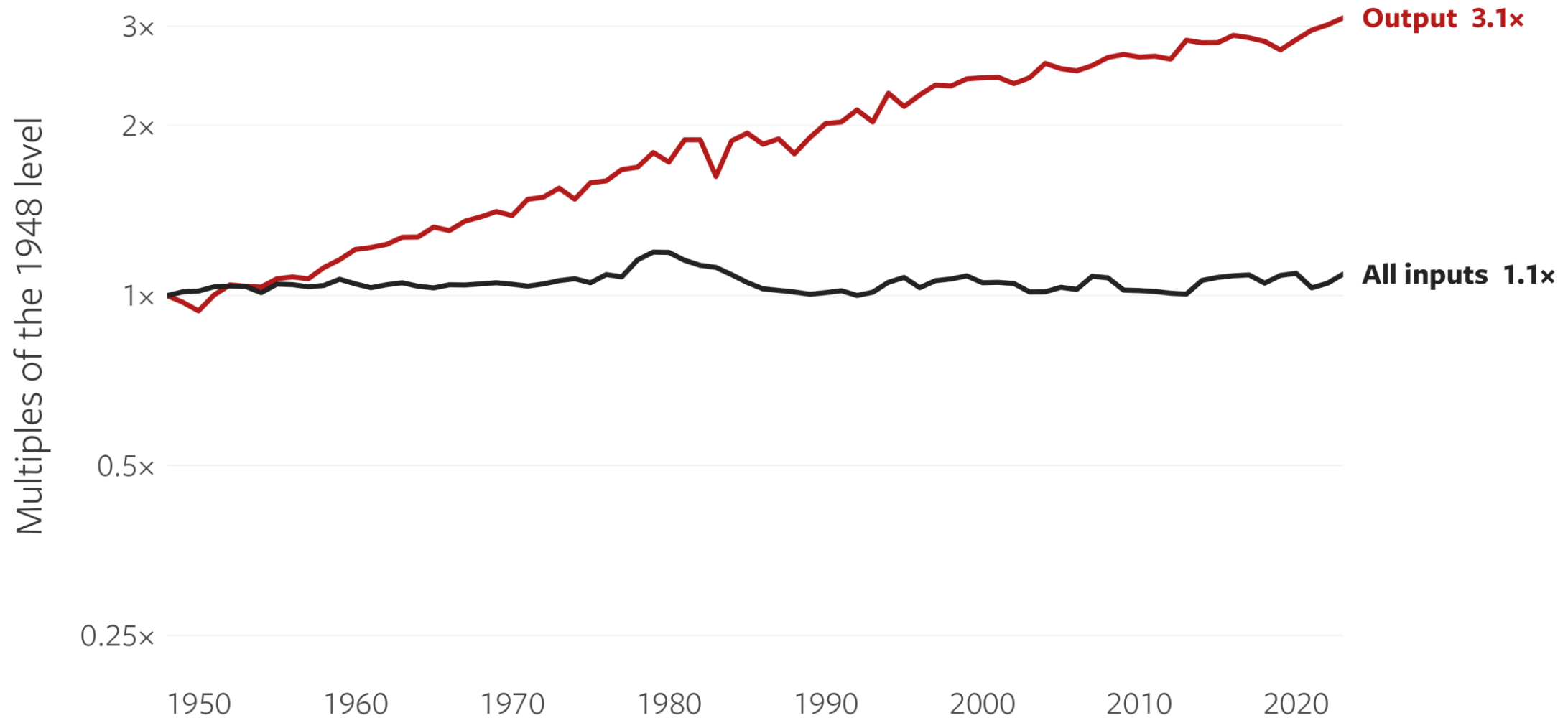


# US agricultural output tripled since the 1950s



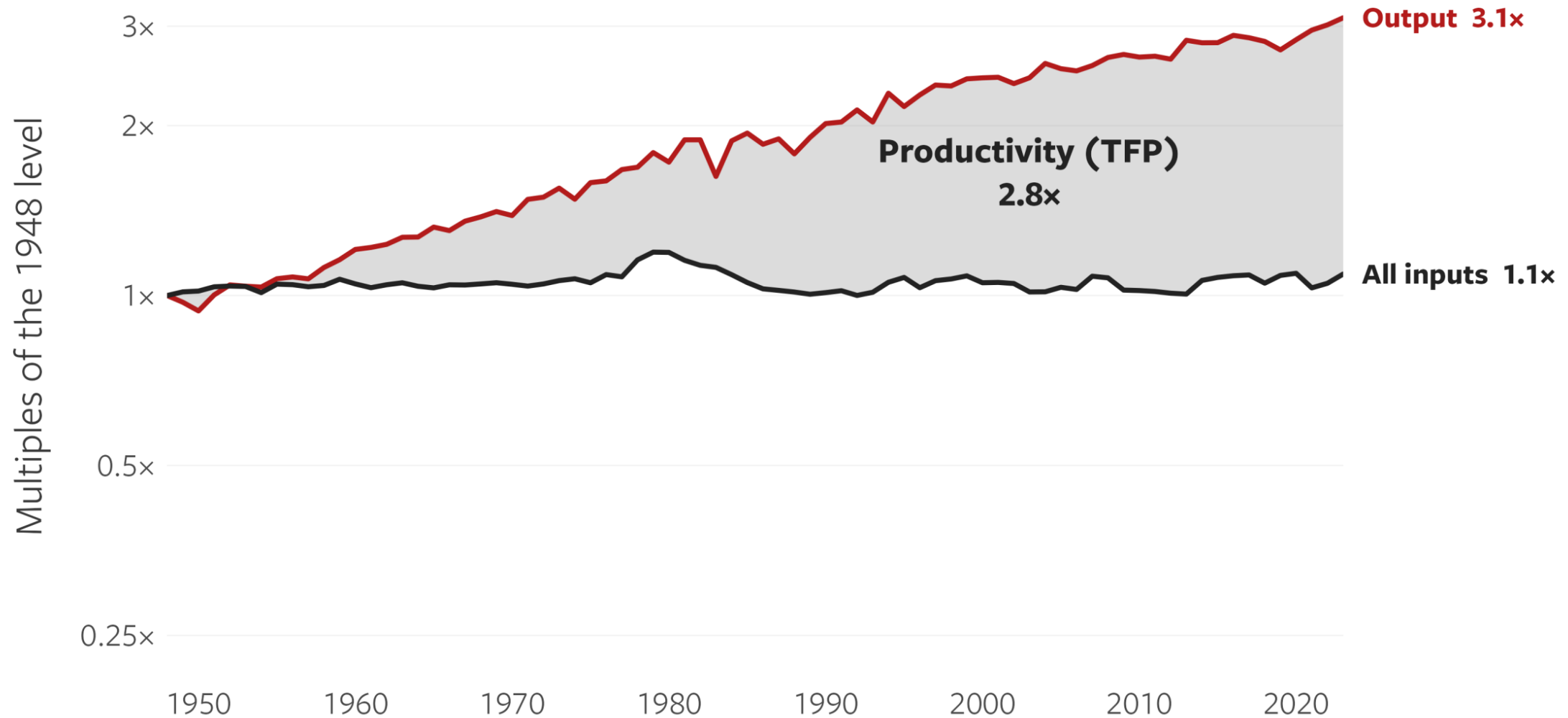
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

# Output tripled with the same inputs



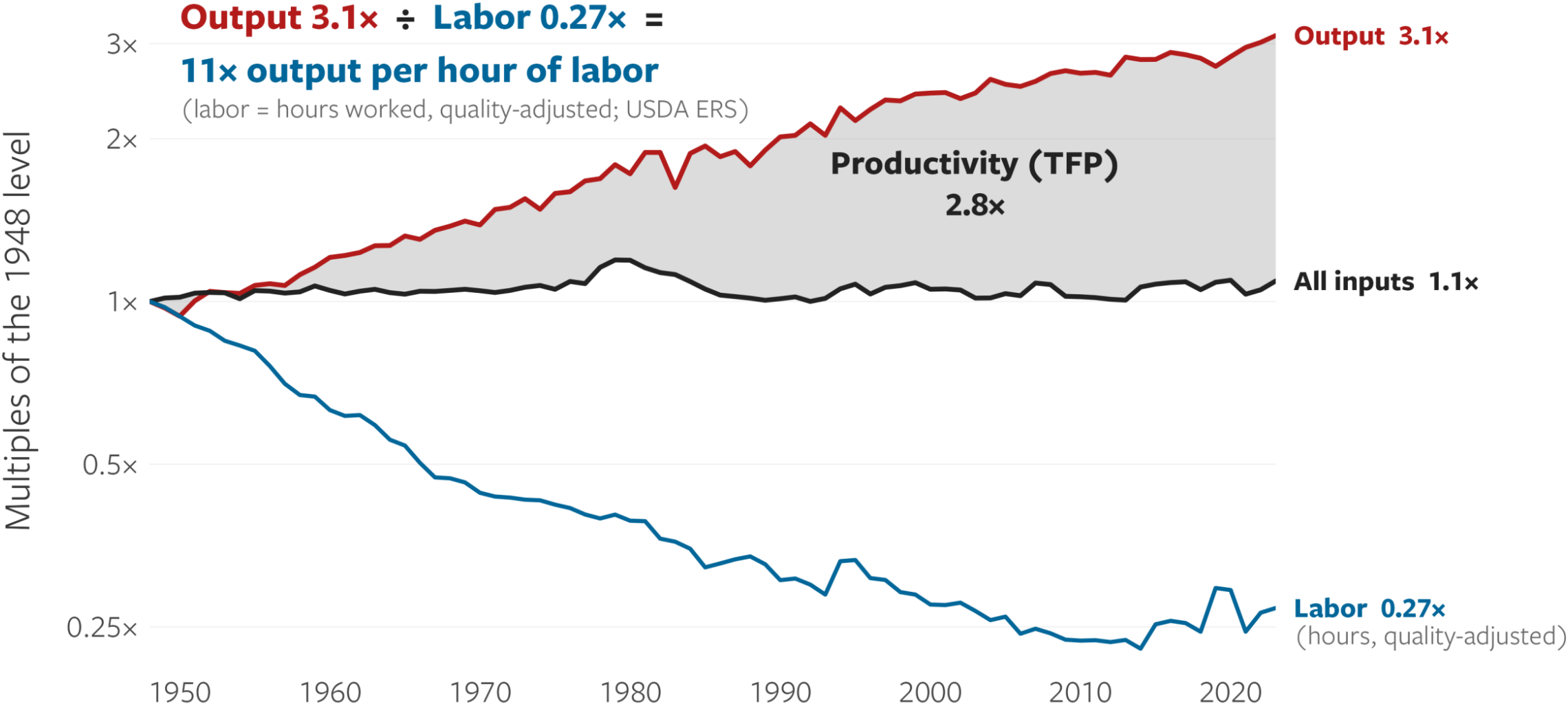
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

# The gap between output and inputs is productivity



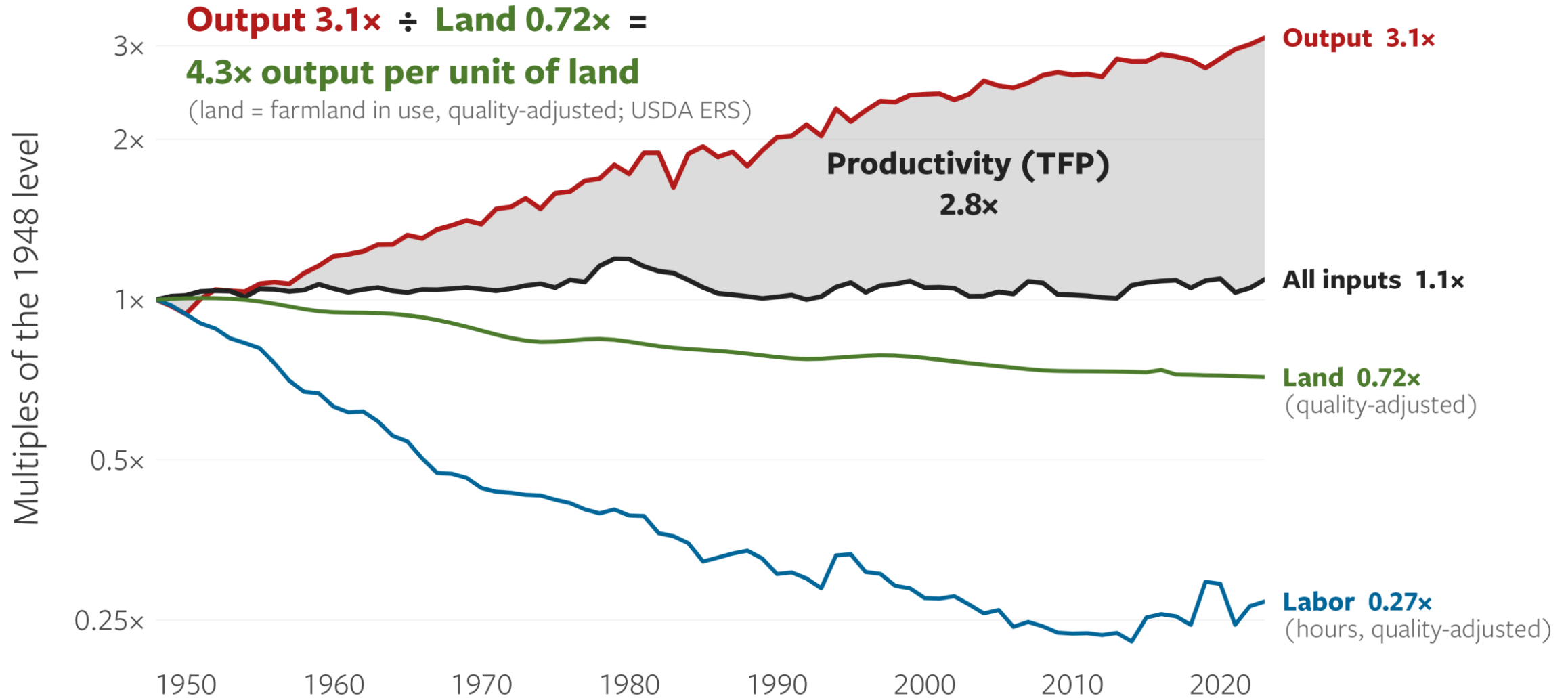
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

# Good for farmers: 11x the output per hour of labor



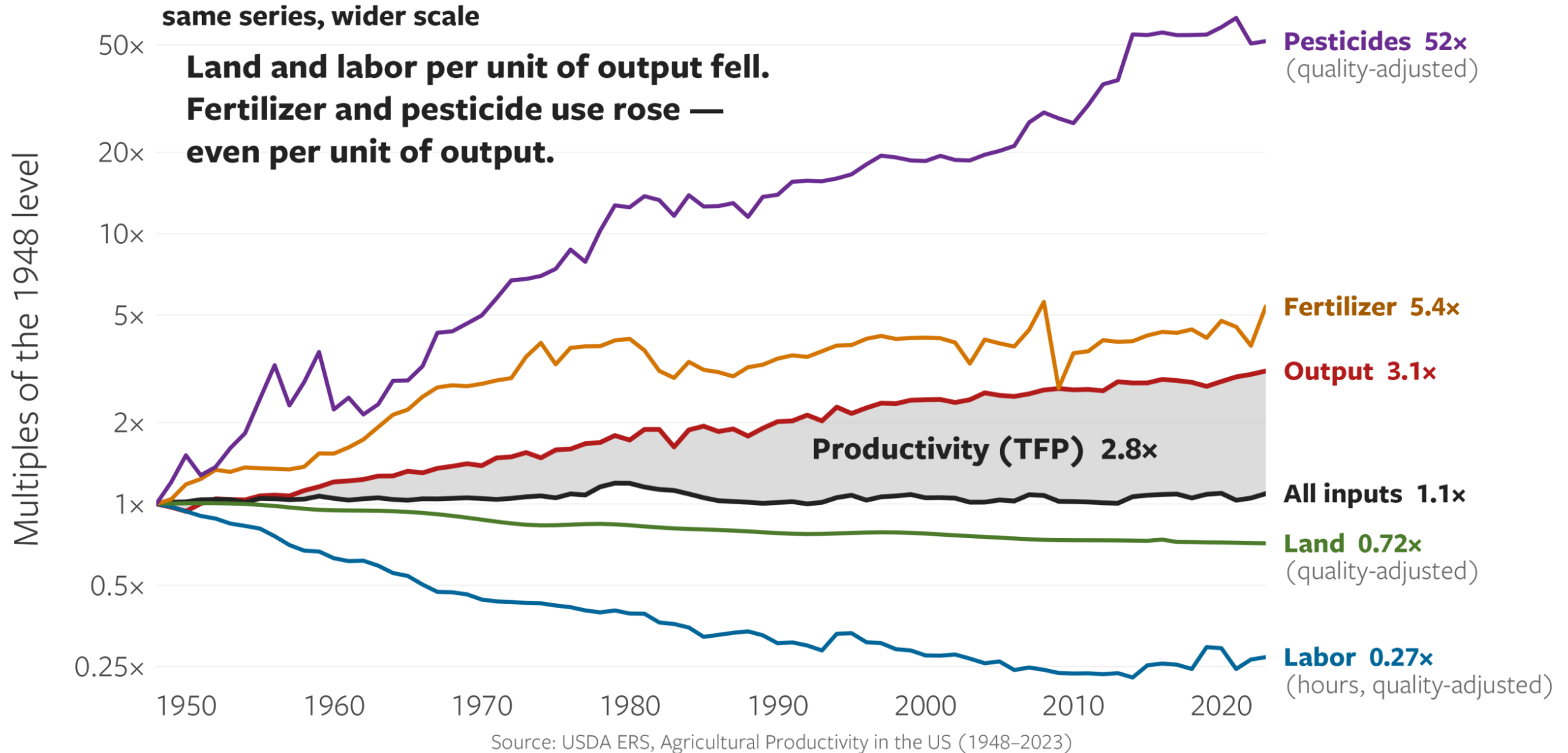
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

# Good for the environment: 4.3× the output per acre

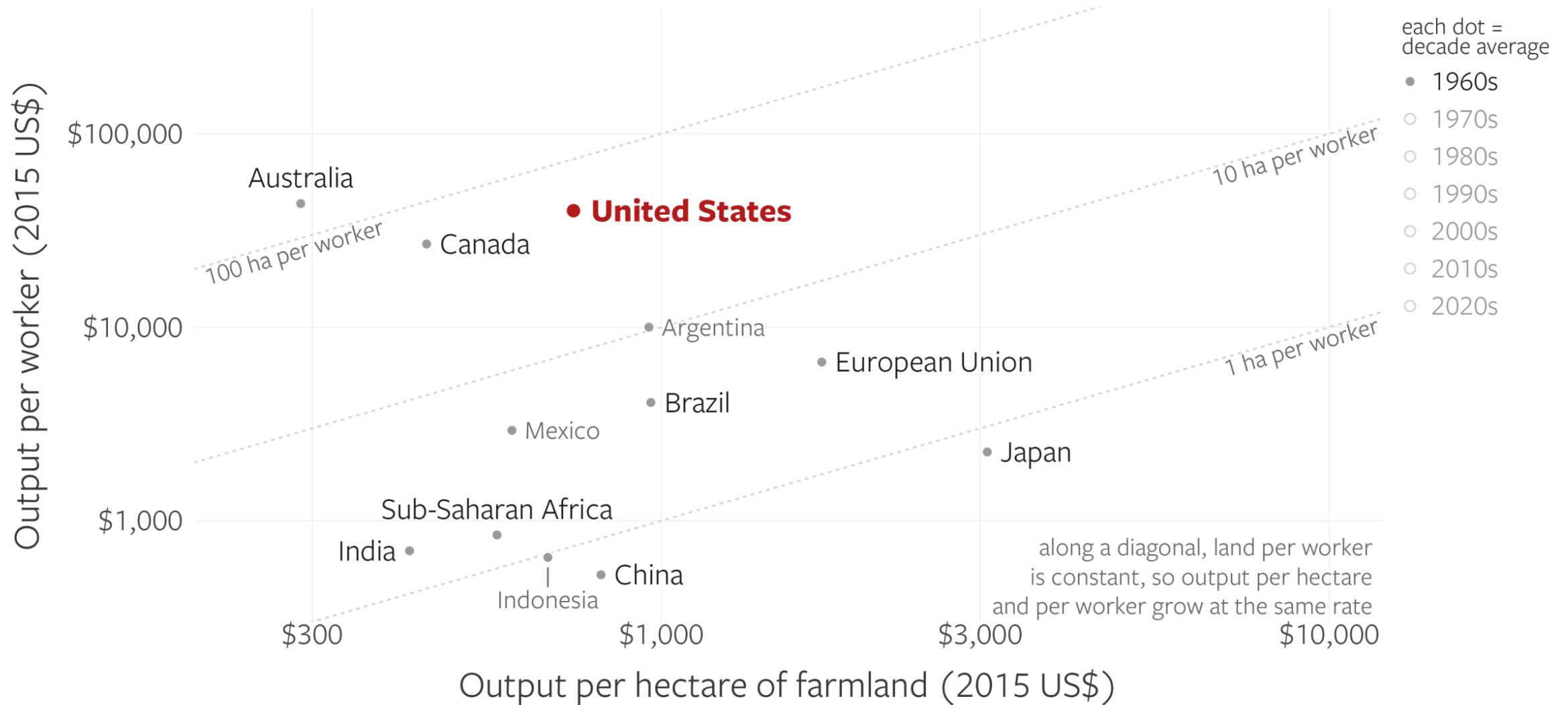


Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

# But fertilizer and pesticide use grew faster than output

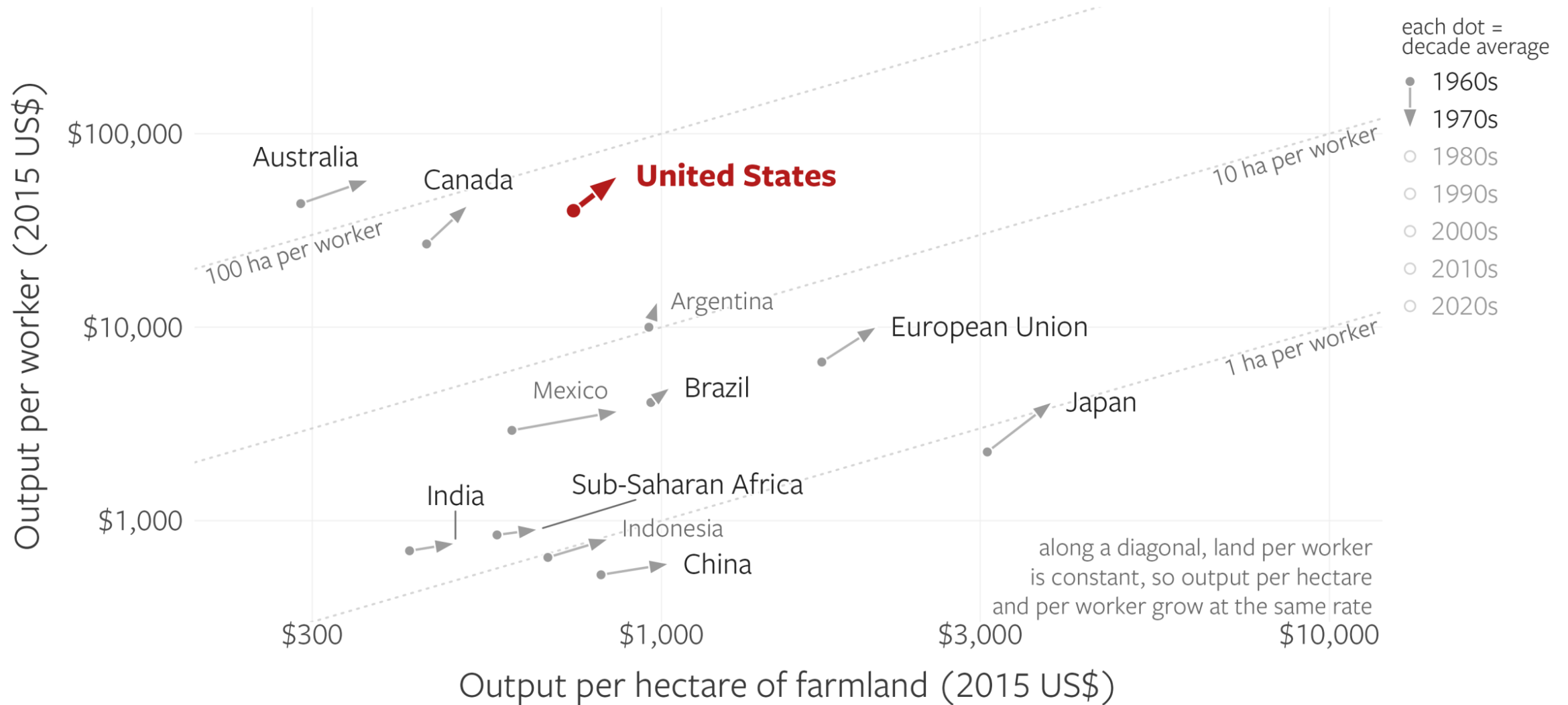


# 1960s



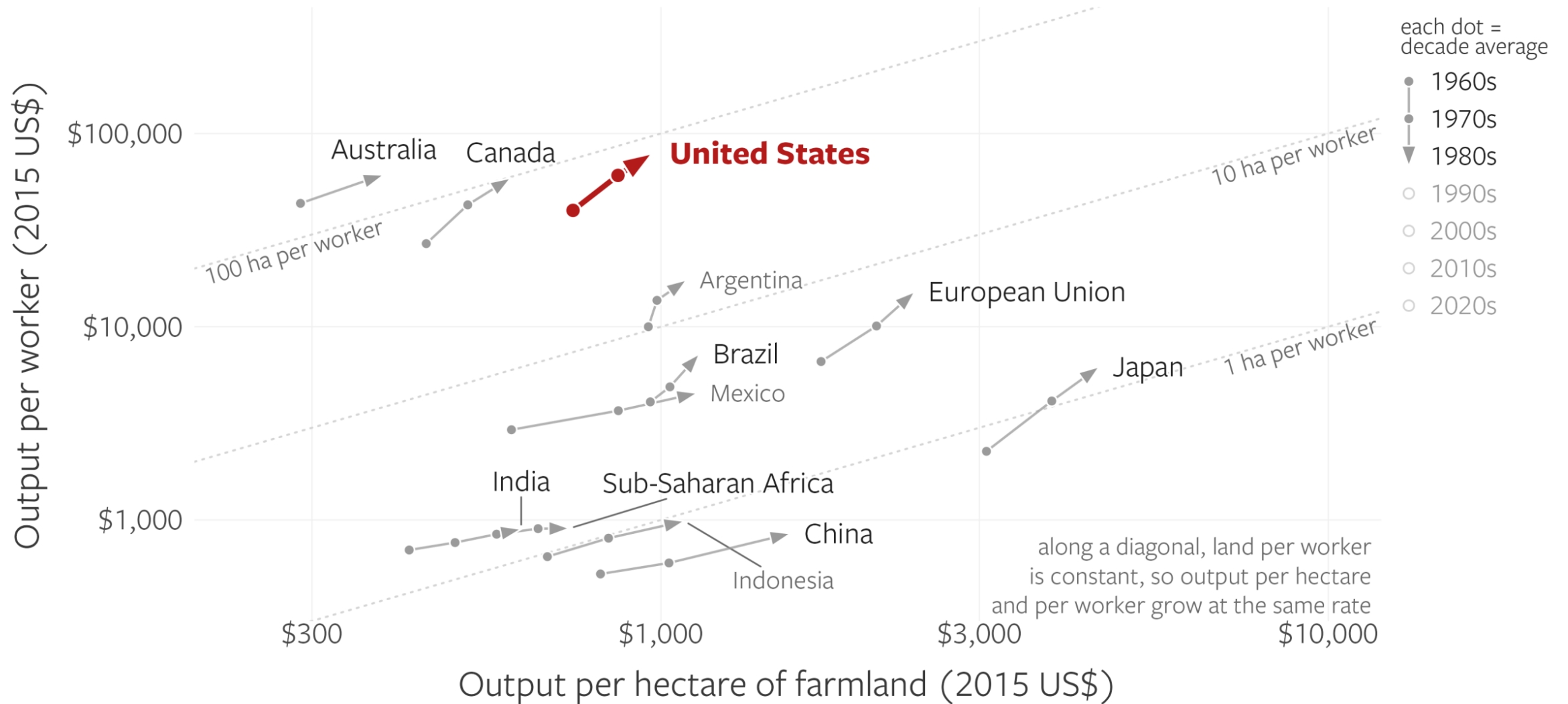
Source: USDA ERS, International Agricultural Productivity (1961–2023); points are decade averages (2020s = 2020–23); dotted diagonals: constant land per worker.

# 1970s



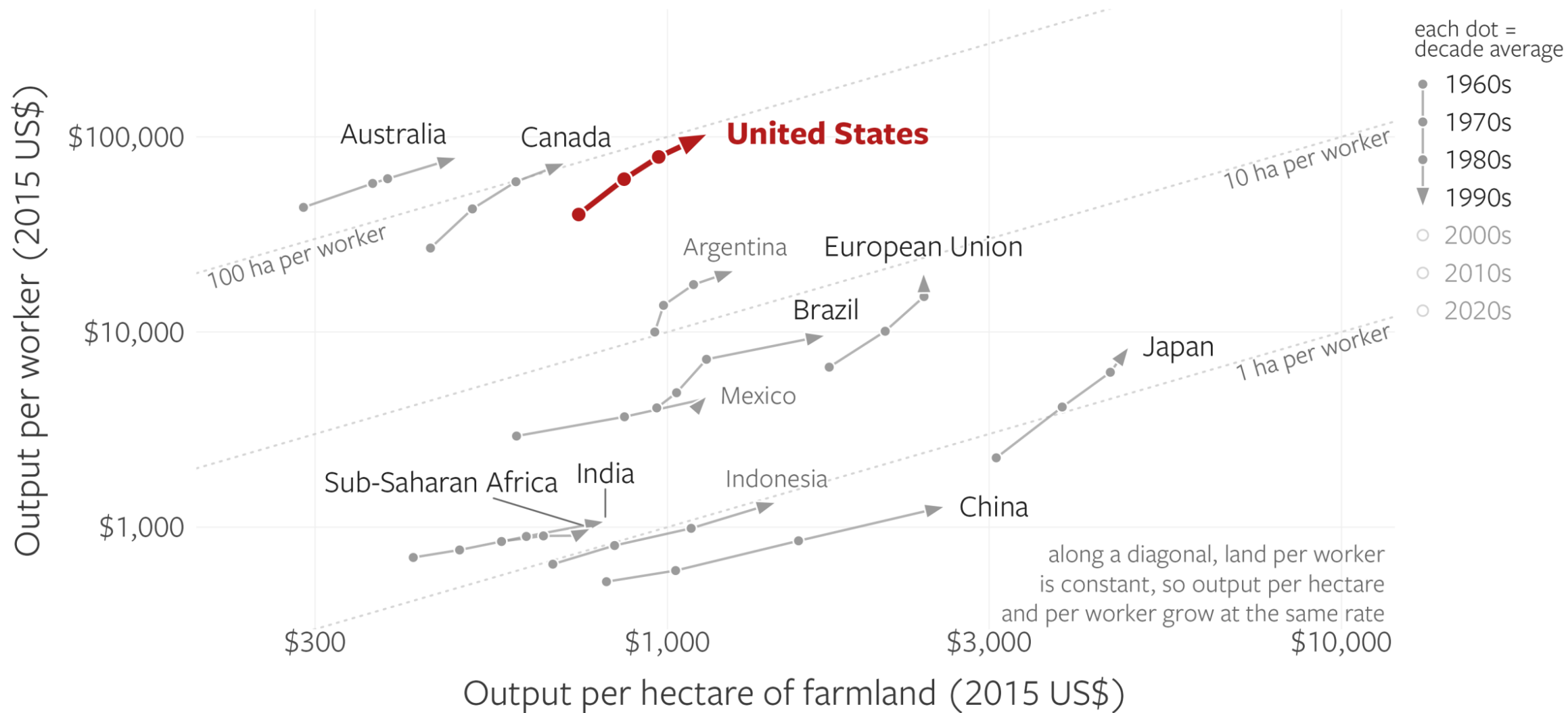
Source: USDA ERS, International Agricultural Productivity (1961–2023); points are decade averages (2020s = 2020–23); dotted diagonals: constant land per worker.

# 1980s



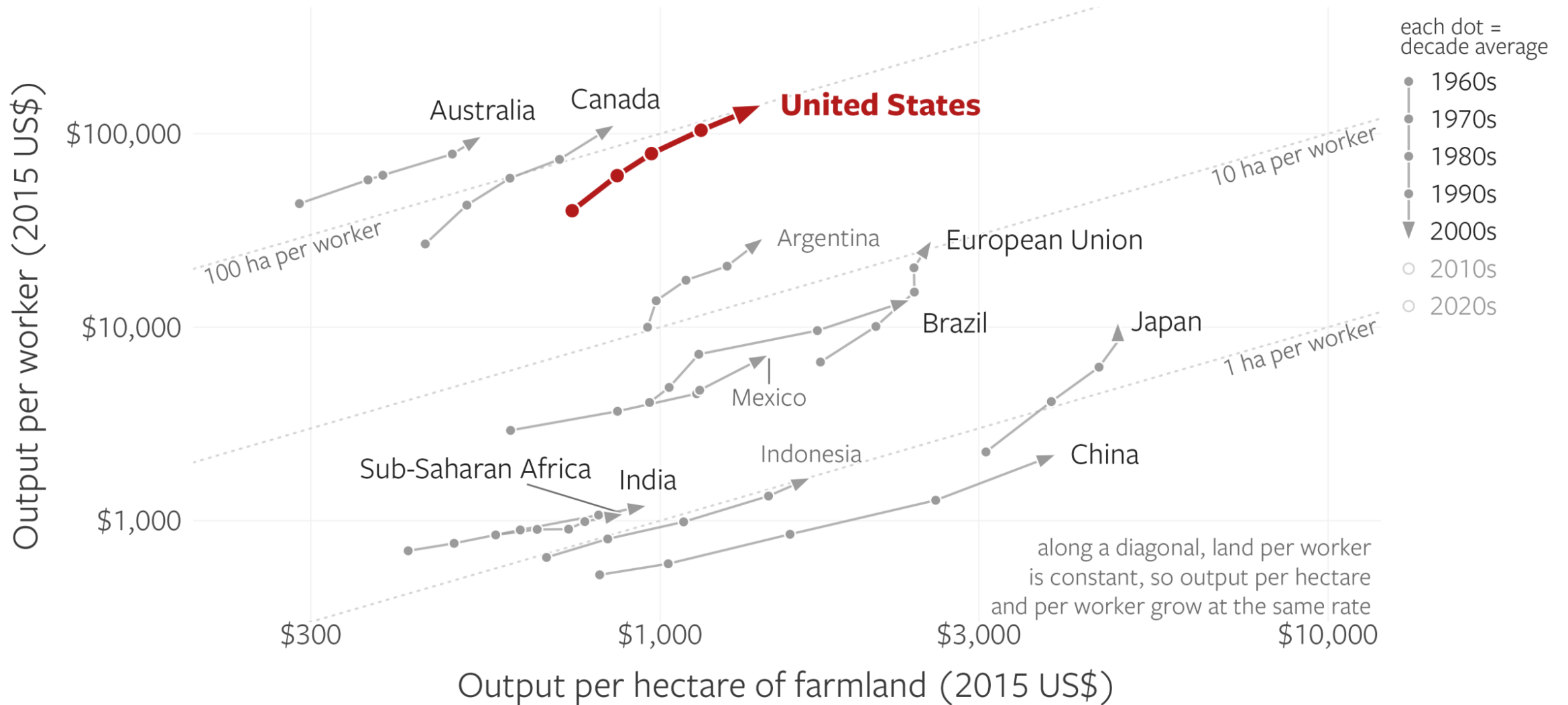
Source: USDA ERS, International Agricultural Productivity (1961–2023); points are decade averages (2020s = 2020–23); dotted diagonals: constant land per worker.

# 1990s



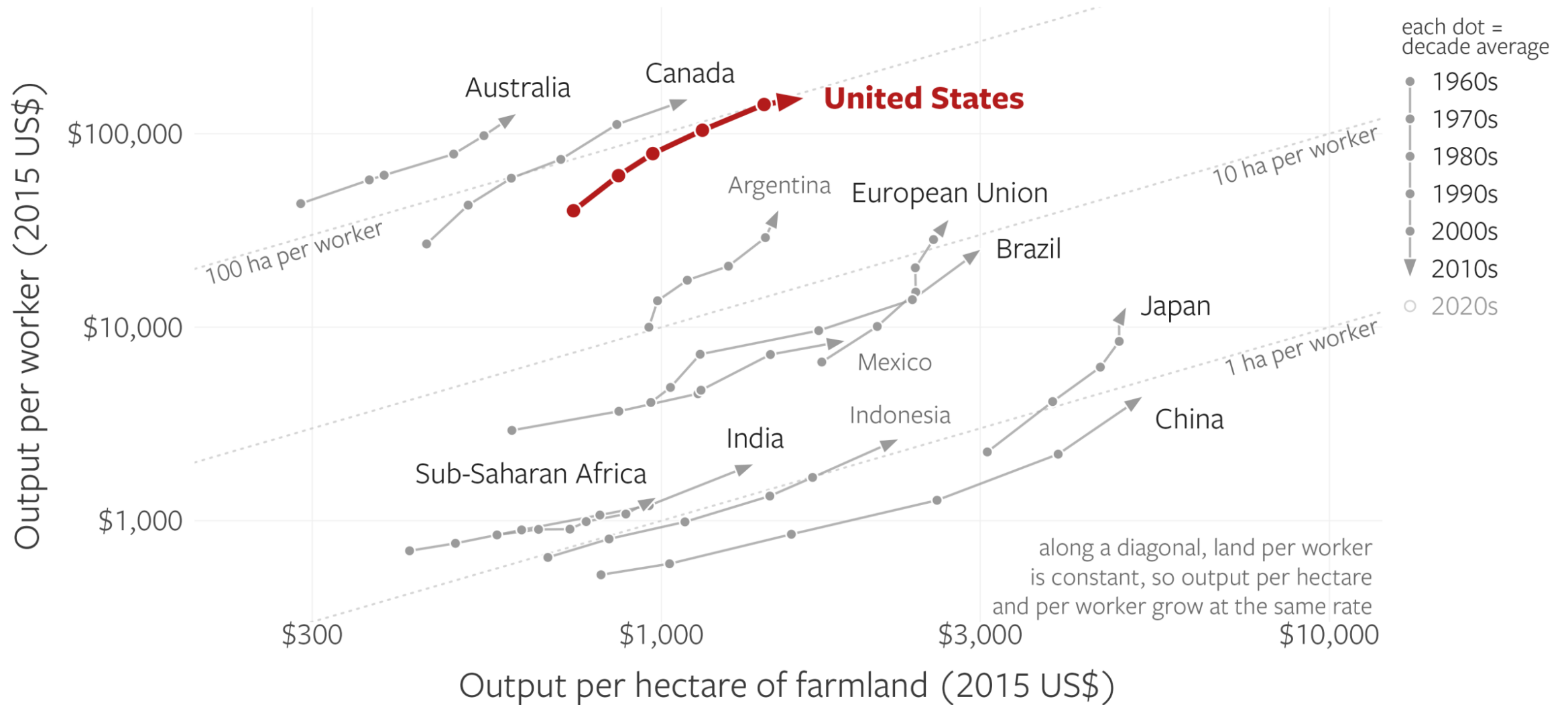
Source: USDA ERS, International Agricultural Productivity (1961–2023); points are decade averages (2020s = 2020–23); dotted diagonals: constant land per worker.

# 2000s



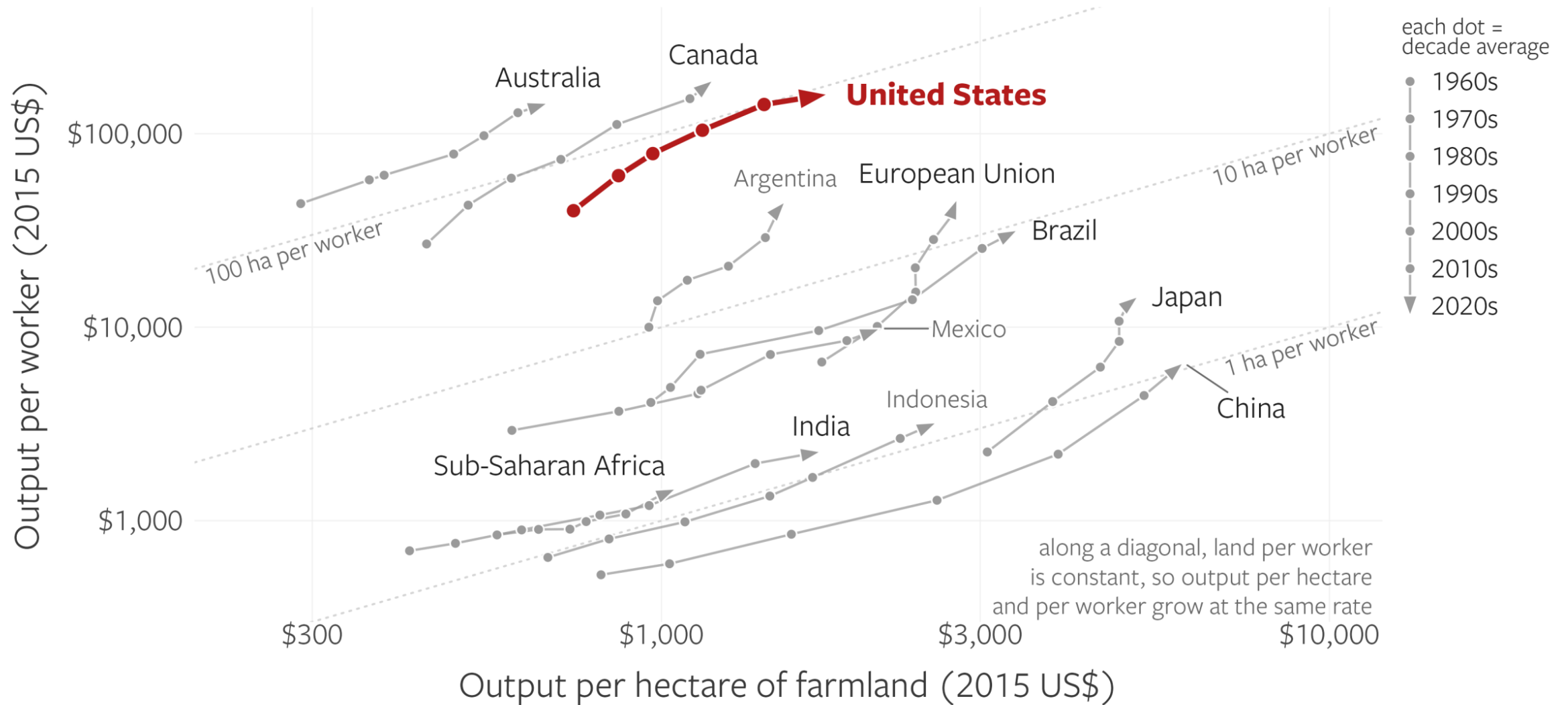
Source: USDA ERS, International Agricultural Productivity (1961–2023); points are decade averages (2020s = 2020–23); dotted diagonals: constant land per worker.

# 2010s



Source: USDA ERS, International Agricultural Productivity (1961–2023); points are decade averages (2020s = 2020–23); dotted diagonals: constant land per worker.

# 2020S



Source: USDA ERS, International Agricultural Productivity (1961–2023); points are decade averages (2020s = 2020–23); dotted diagonals: constant land per worker.

# Key messages of this talk

**1**

**Productivity growth is slowing.**

**2**

**The main driver is slowing R&D —  
but increasingly it's climate.**

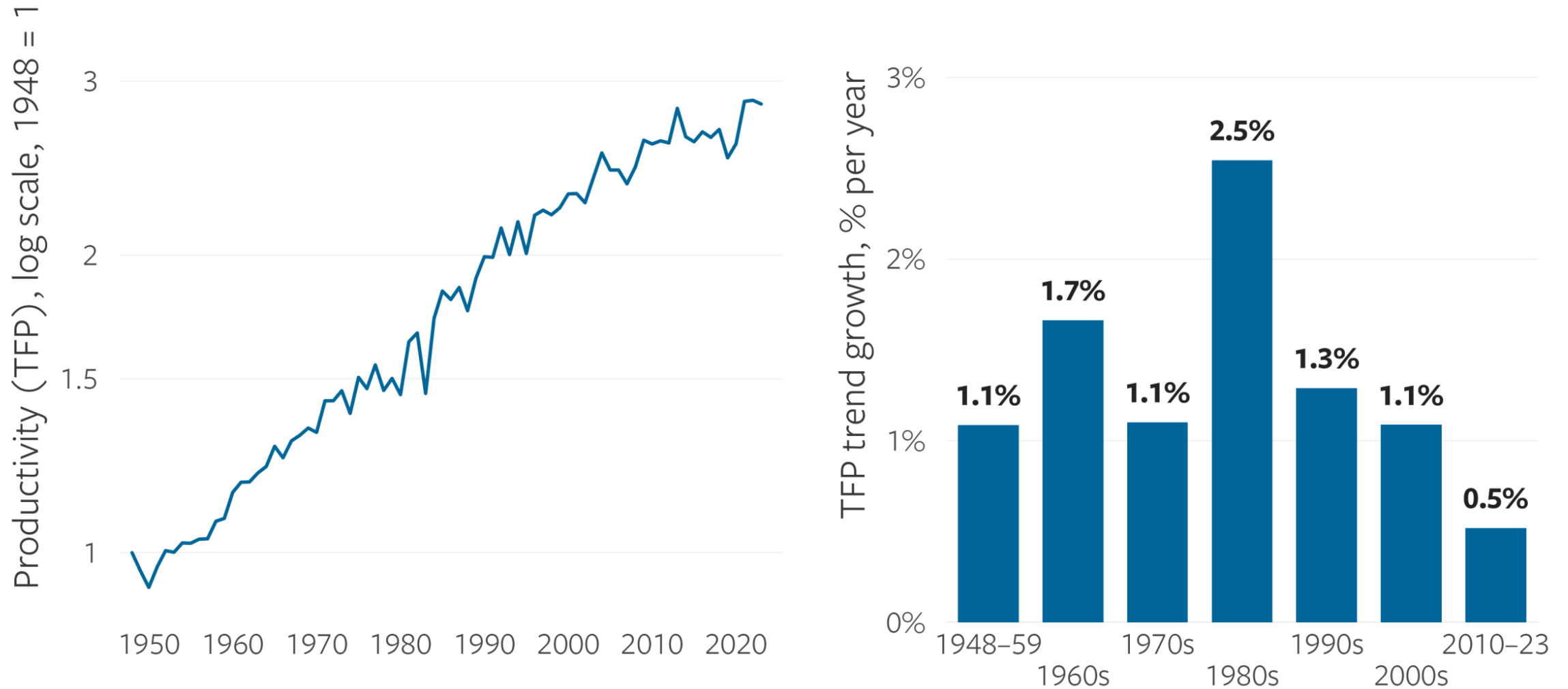
**3**

**Maintaining productivity growth  
requires a historical pivot.**

# Message 1

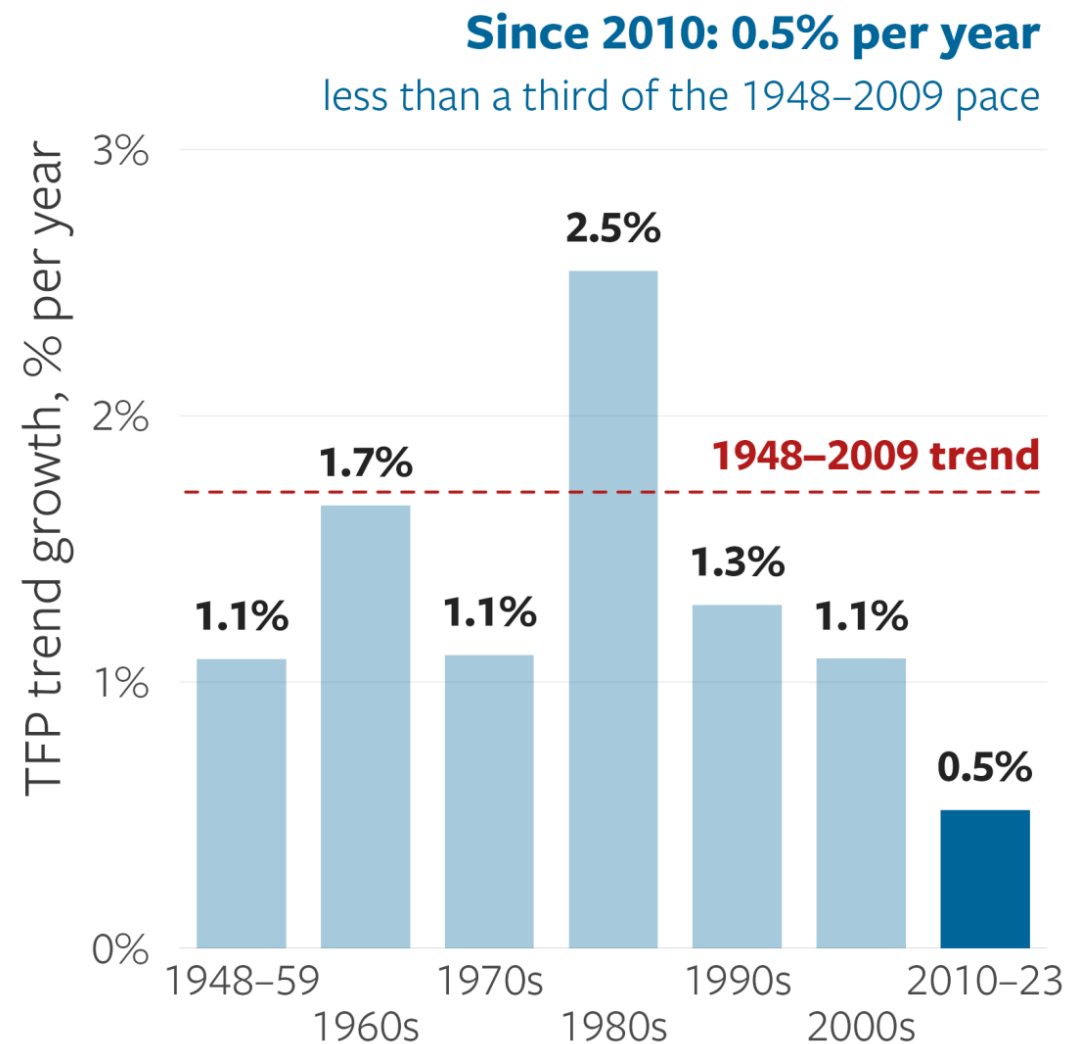
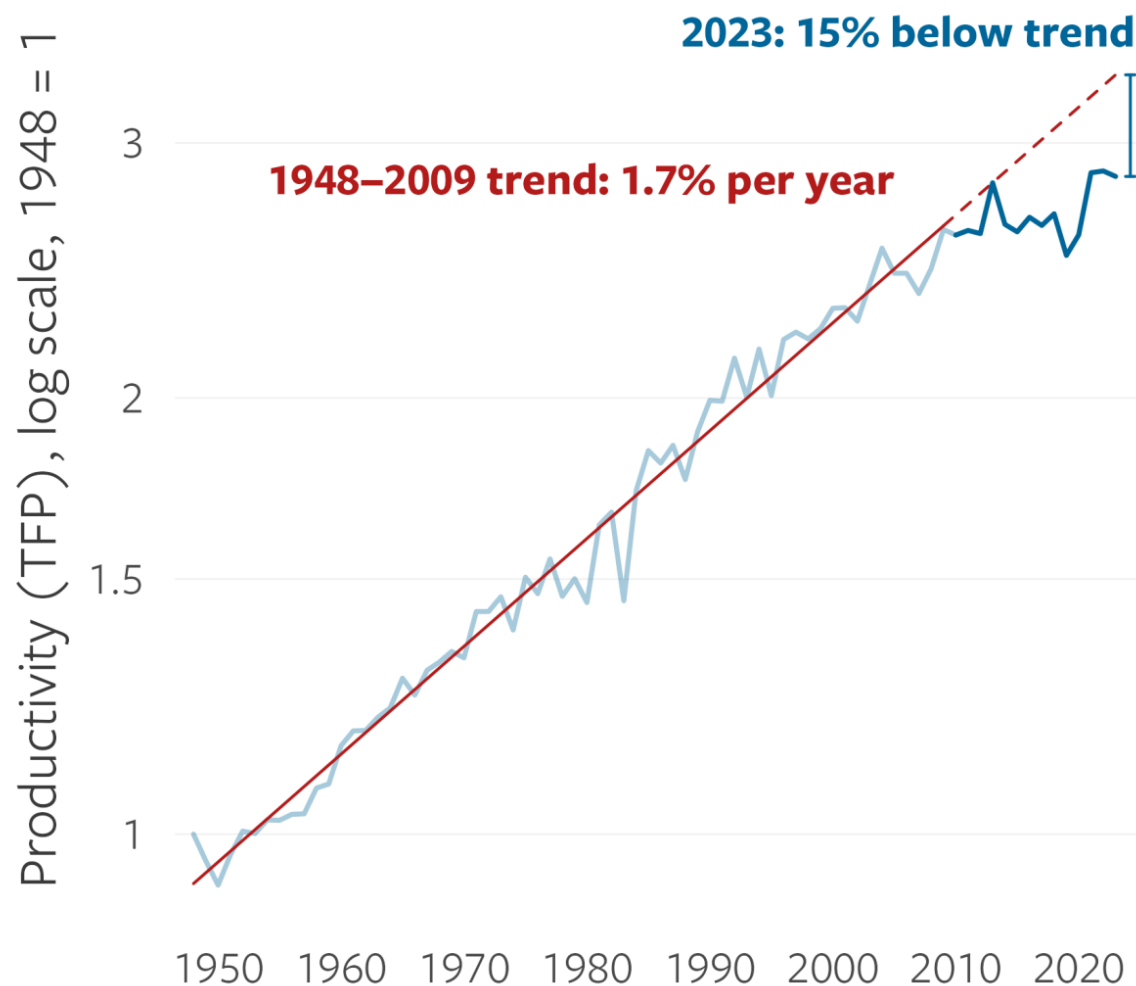
- 1 Productivity growth is slowing.**
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.**

# Productivity growth, decade by decade



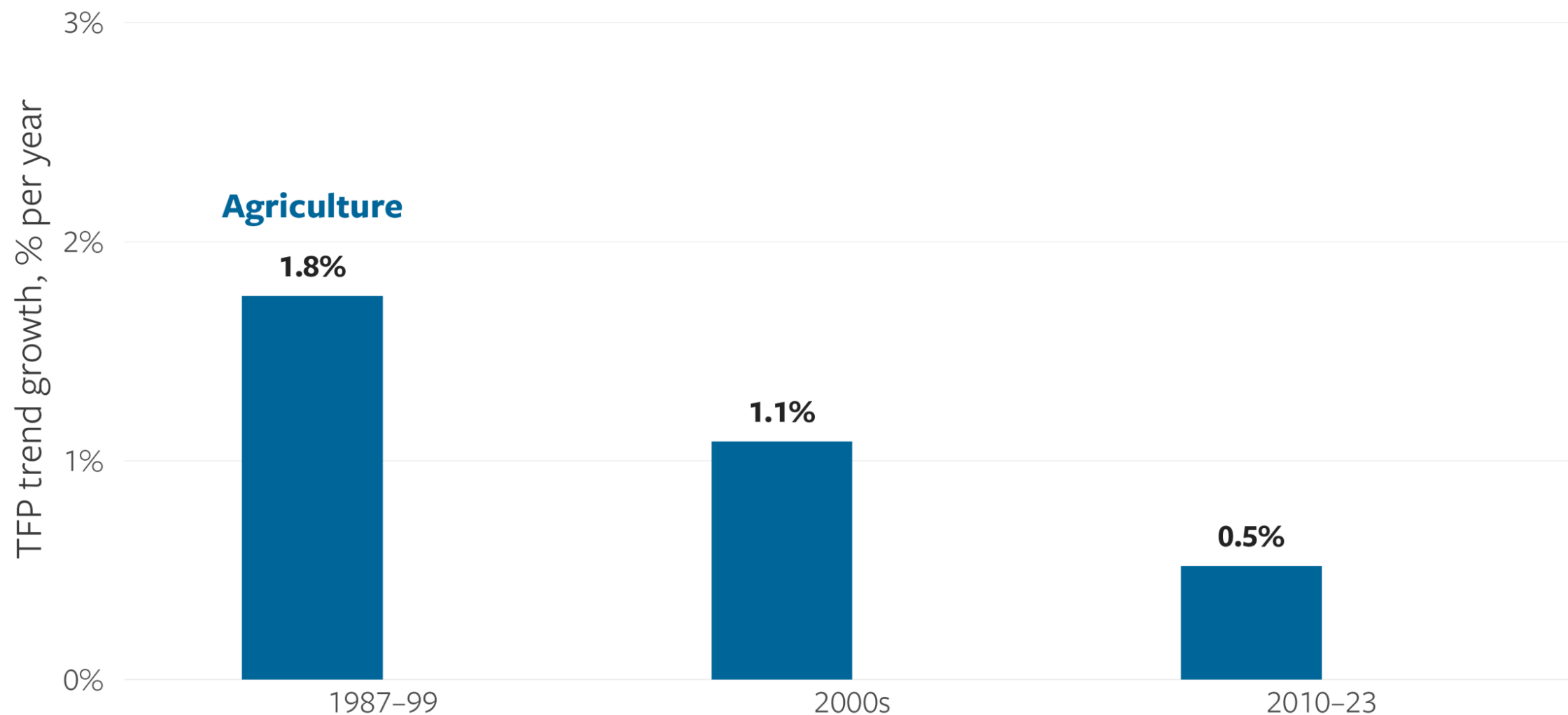
Source: USDA ERS, Agricultural Productivity in the US (1948-2023). Bars: trend growth of log TFP within each window.

# The engine is slowing: productivity is 15% below its postwar trend



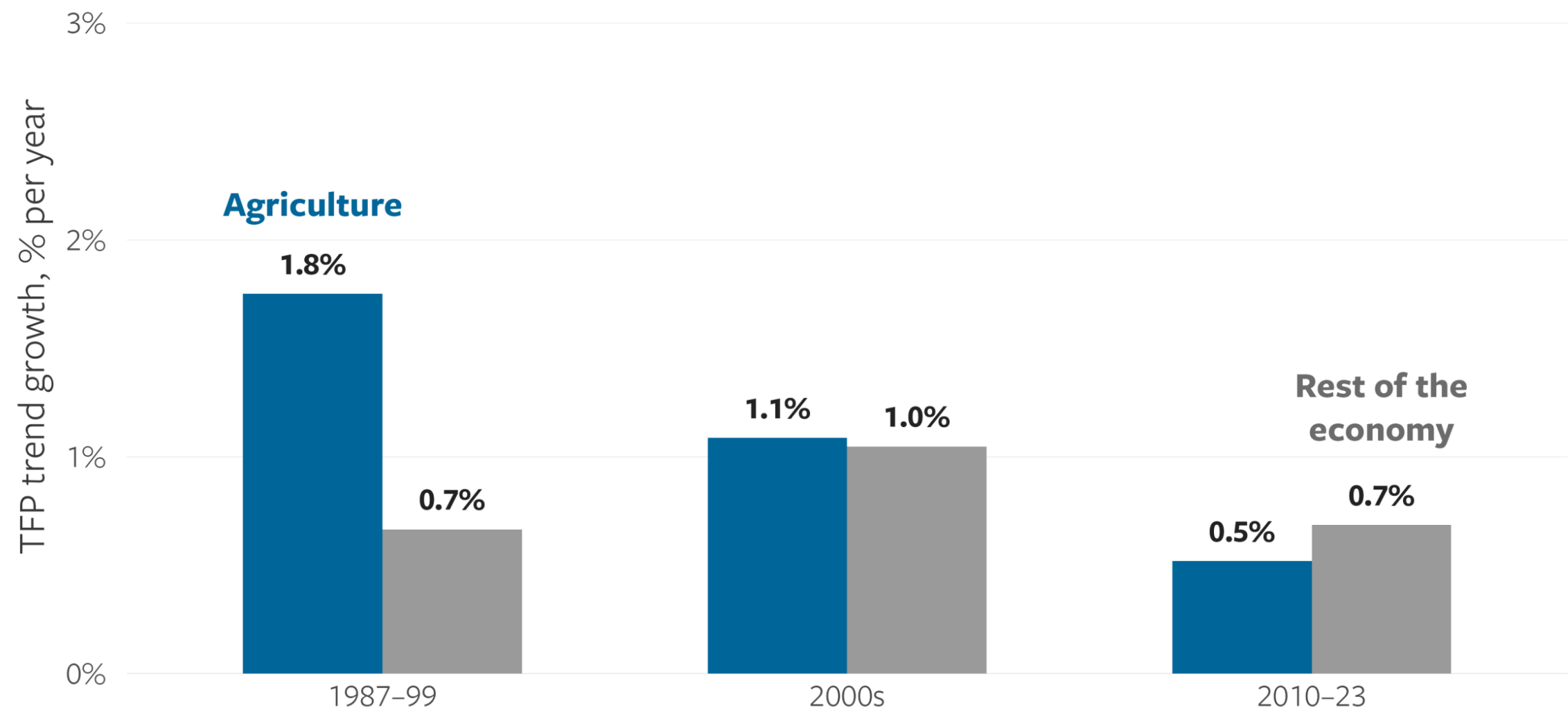
Source: USDA ERS, Agricultural Productivity in the US (1948–2023). Bars: trend growth of log TFP within each window.

# Productivity growth by decade: agriculture



Source: USDA ERS (farm TFP); BLS (TFP, private nonfarm business sector). Trend growth within each window.

# Agriculture led the economy for two decades; since 2010 it has fallen behind



Source: USDA ERS (farm TFP); BLS (TFP, private nonfarm business sector). Trend growth within each window.

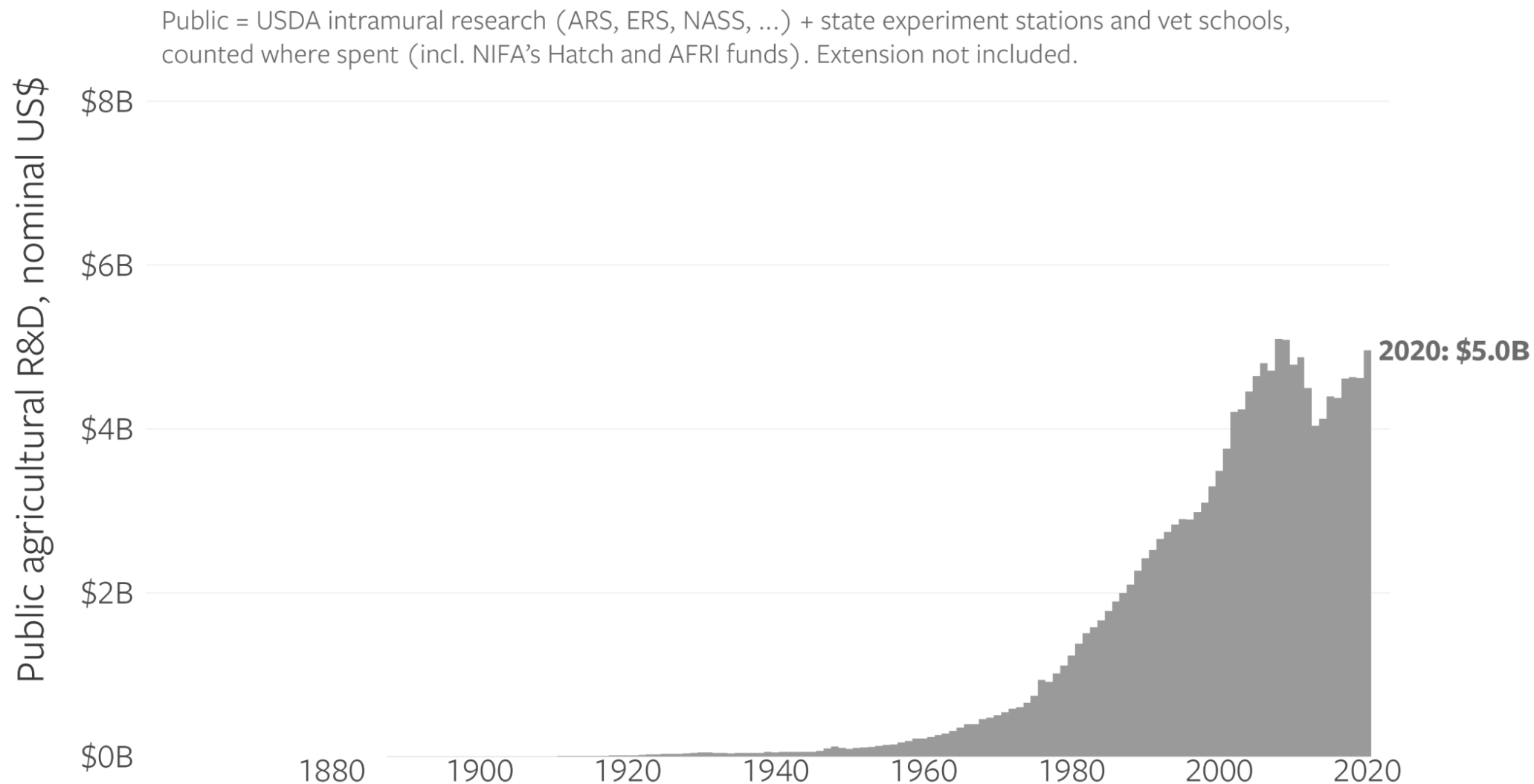
# Message 1 — so why?

- 1 Productivity growth is slowing.**
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.**

## Message 2

- 1 Productivity growth is slowing.
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.

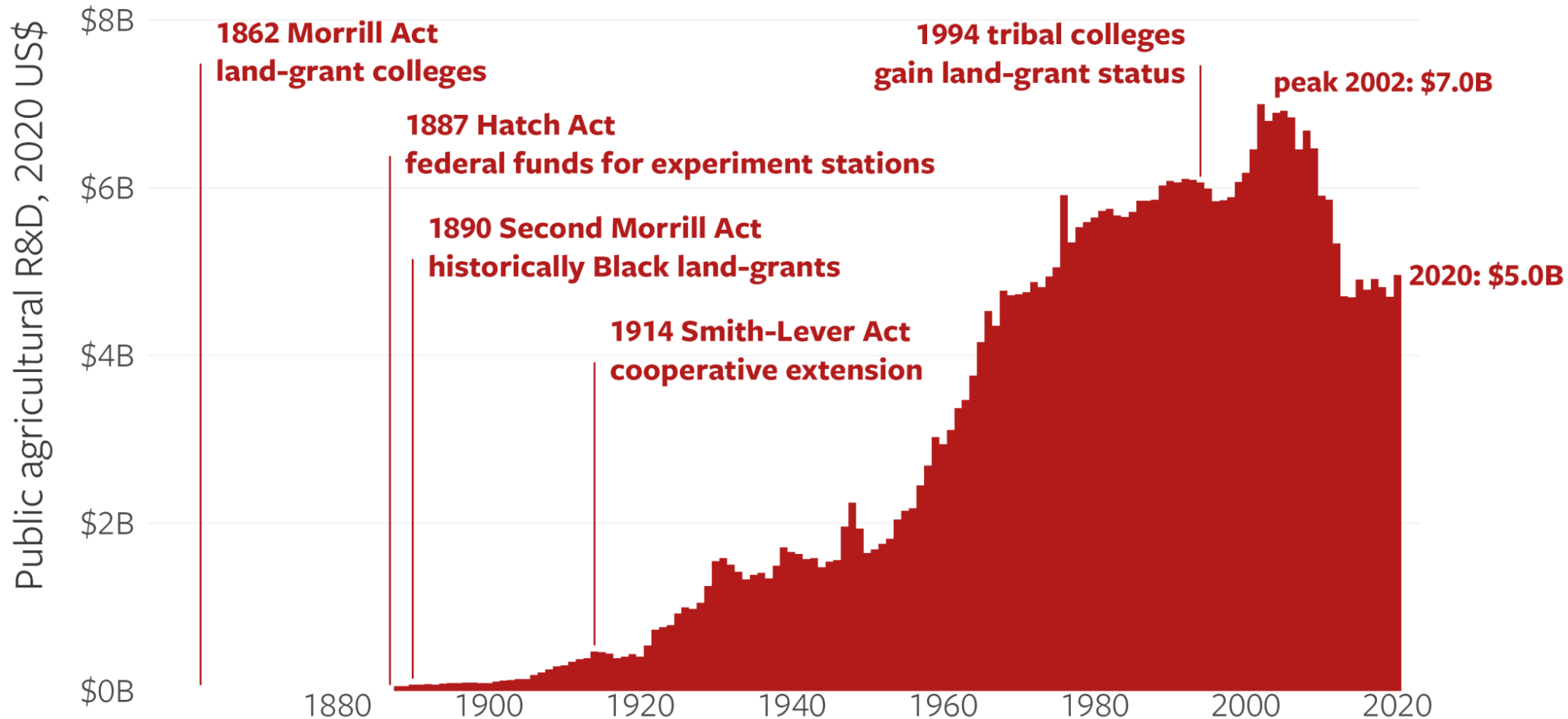
# It looks like we spend more on research every year



Source: USDA ERS long-term public ag R&D series (Huffman & Evenson 2006 to 1950; ERS after), deflated with the NIH BRDPI; replication package of Ortiz-Bobea et al. (PNAS 2025).

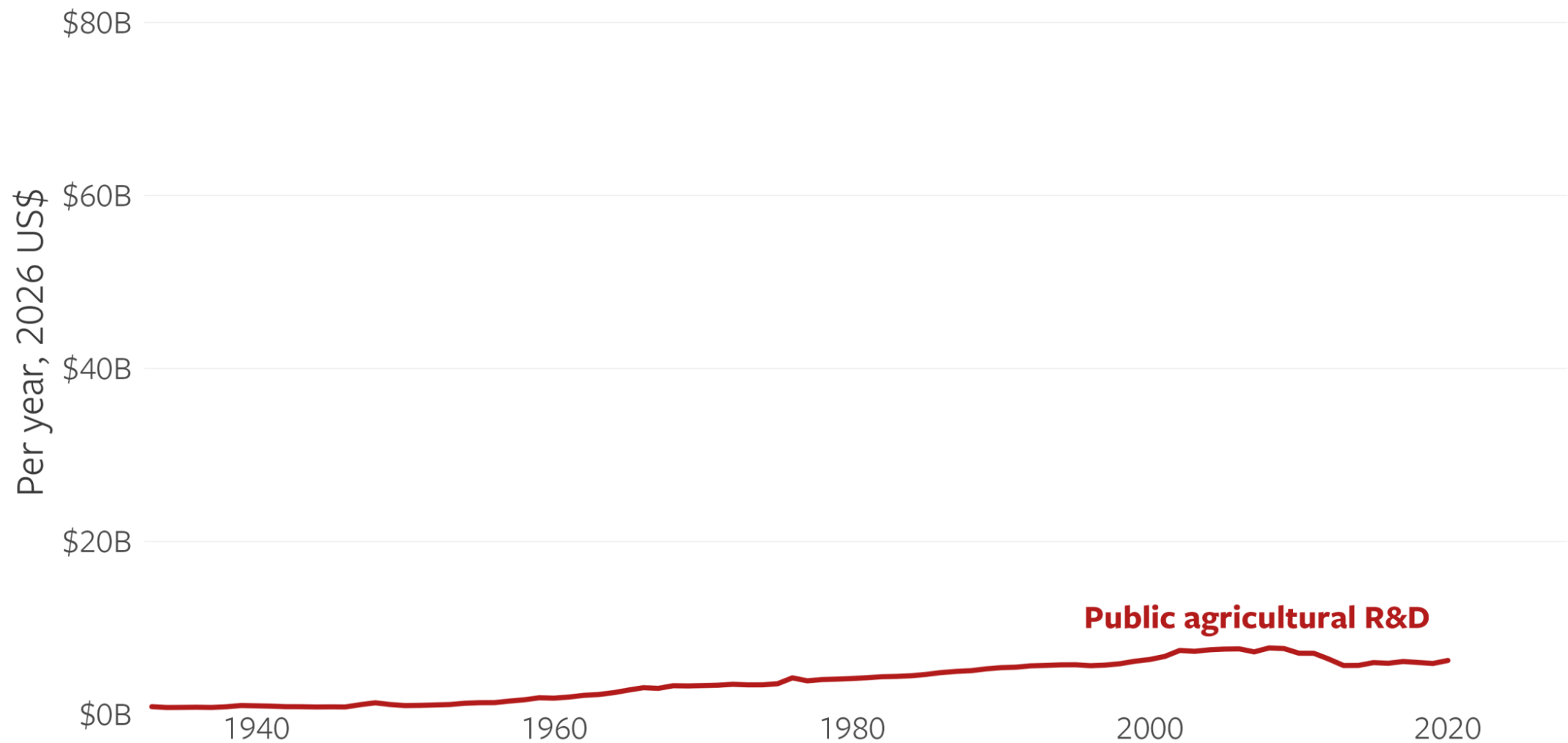
# In real terms, we stopped forty years ago

Public = USDA intramural research (ARS, ERS, NASS, ...) + state experiment stations and vet schools, counted where spent (incl. NIFA's Hatch and AFRI funds). Extension not included.



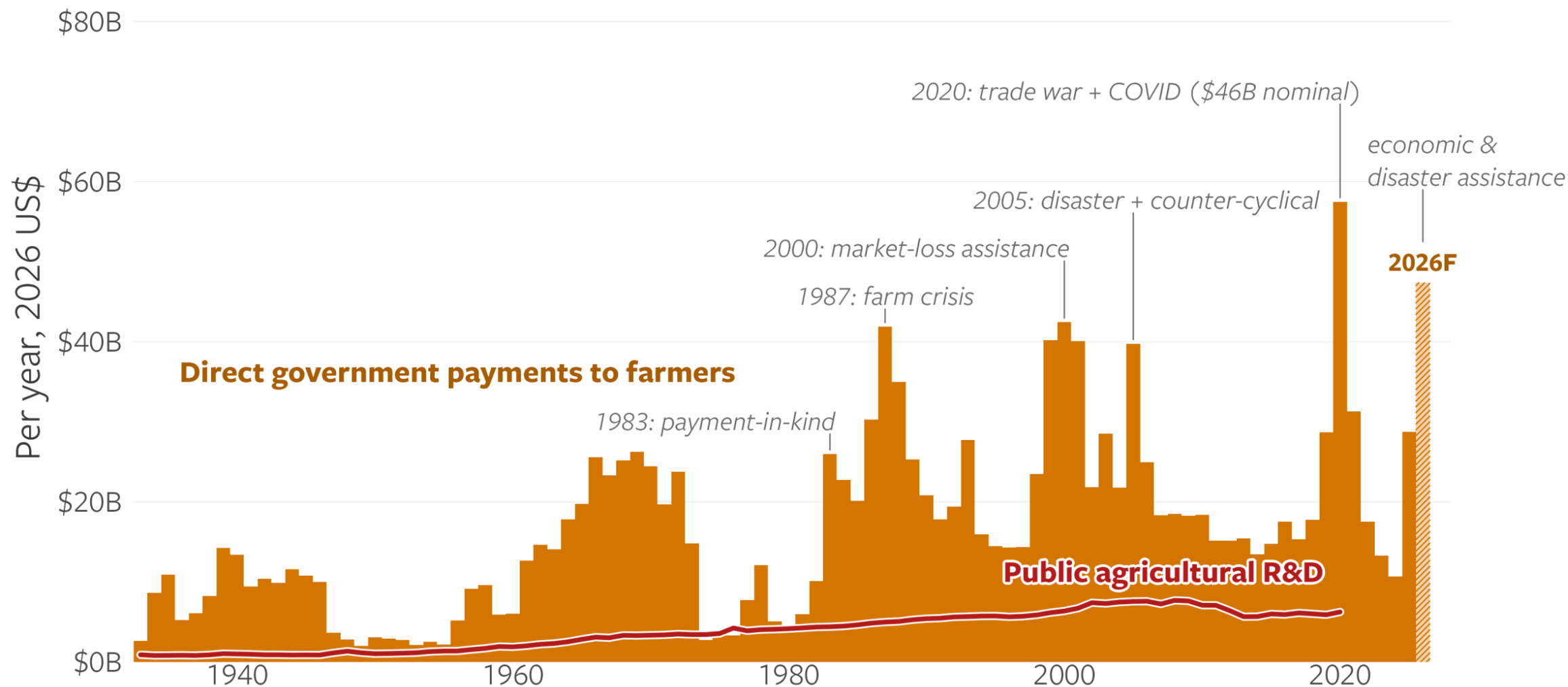
Source: USDA ERS long-term public ag R&D series (Huffman & Evenson 2006 to 1950; ERS after), deflated with the NIH BRDPI; replication package of Ortiz-Bobea et al. (PNAS 2025).

# Public money in agriculture, in today's dollars: research...



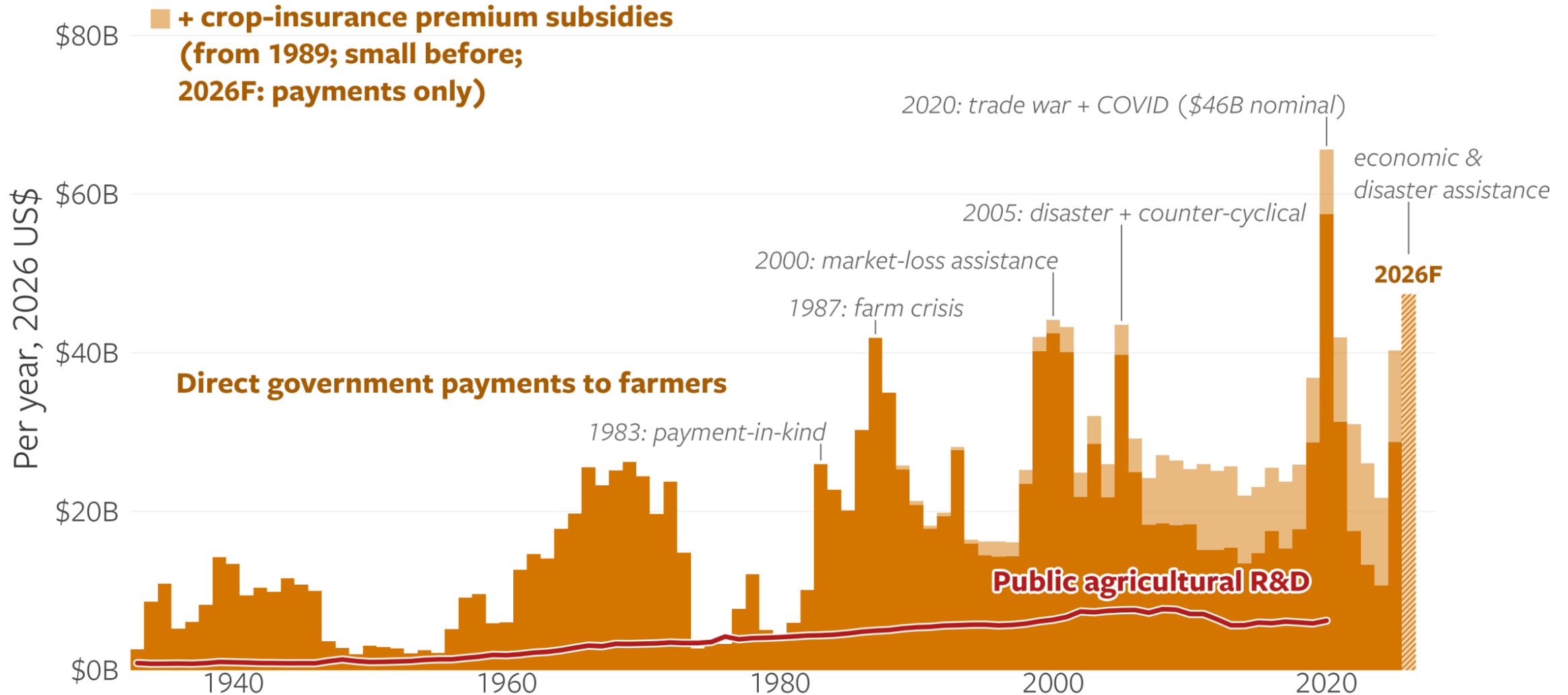
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D. All in 2026 \$ via the GDP deflator.

# ...and direct payments to farmers



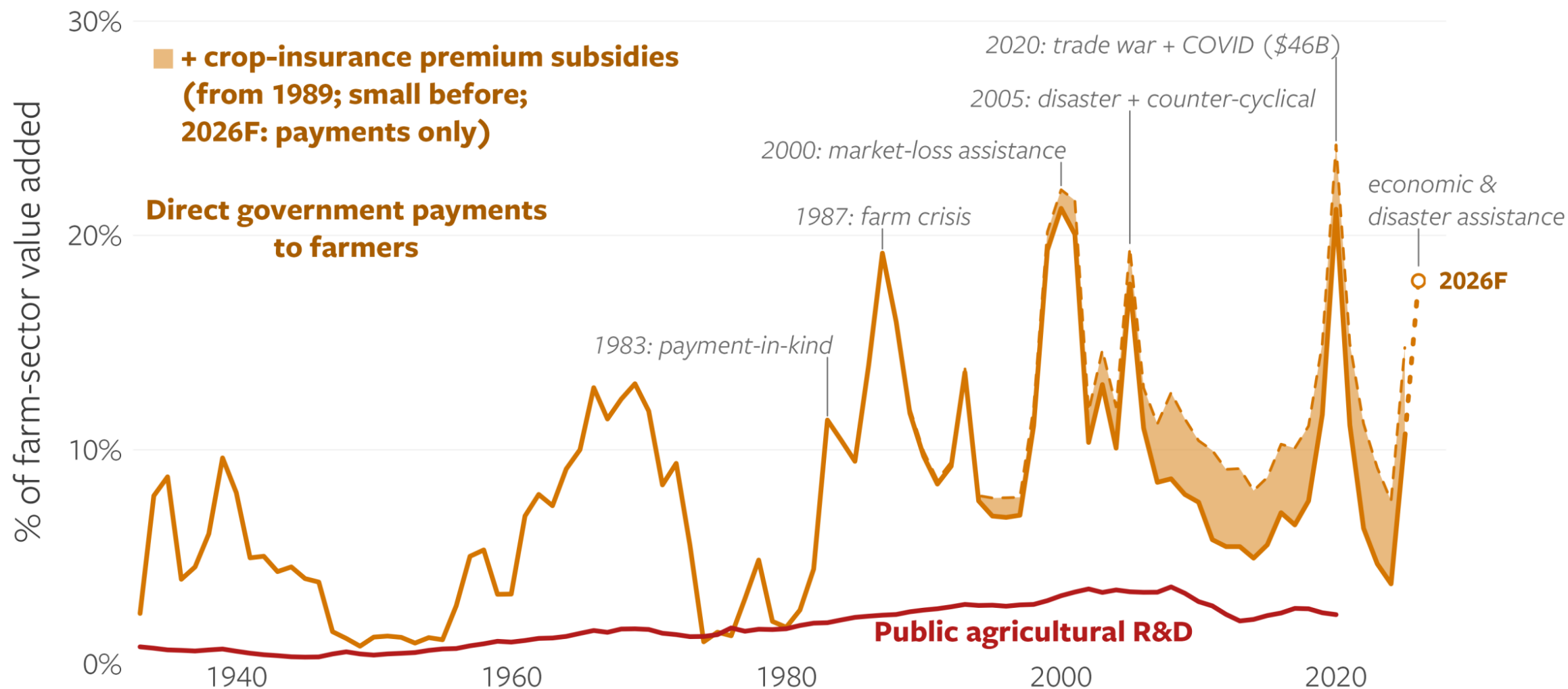
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D. All in 2026 \$ via the GDP deflator.

# ...and crop-insurance premium subsidies



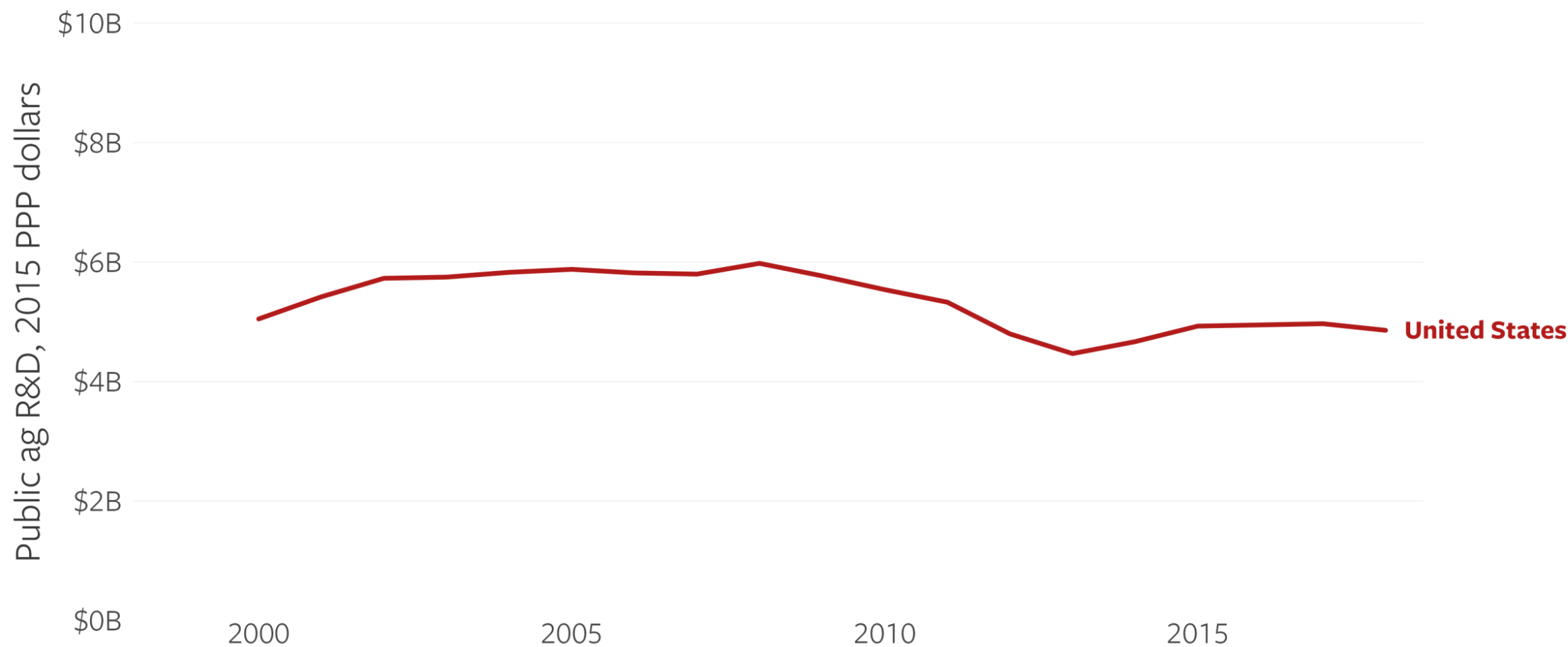
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D. All in 2026 \$ via the GDP deflator.

# As a share of farm GDP



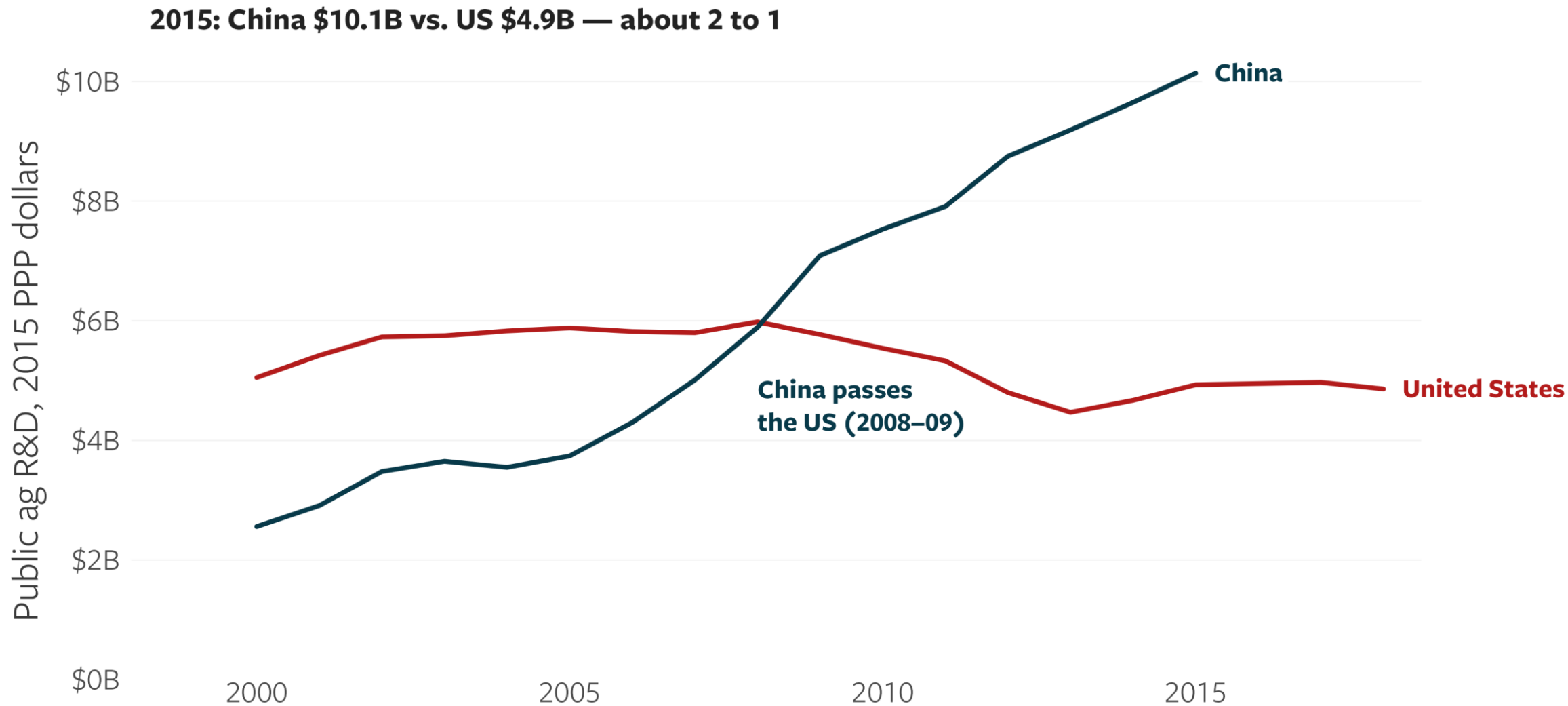
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026 release; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D.

# Public ag R&D: the United States versus others



Source: USDA ERS chart of note (2022) based on ERS, ASTI/IFPRI and World Bank data; values digitized from the published chart.

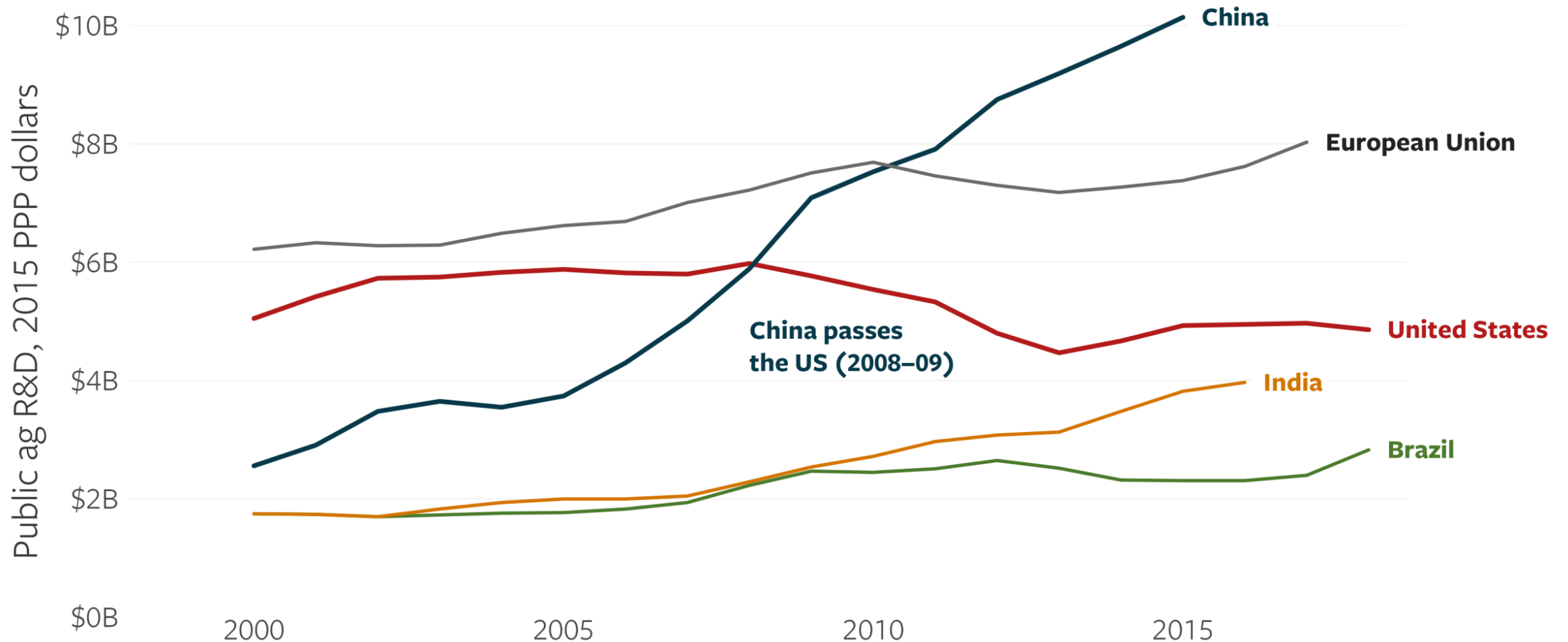
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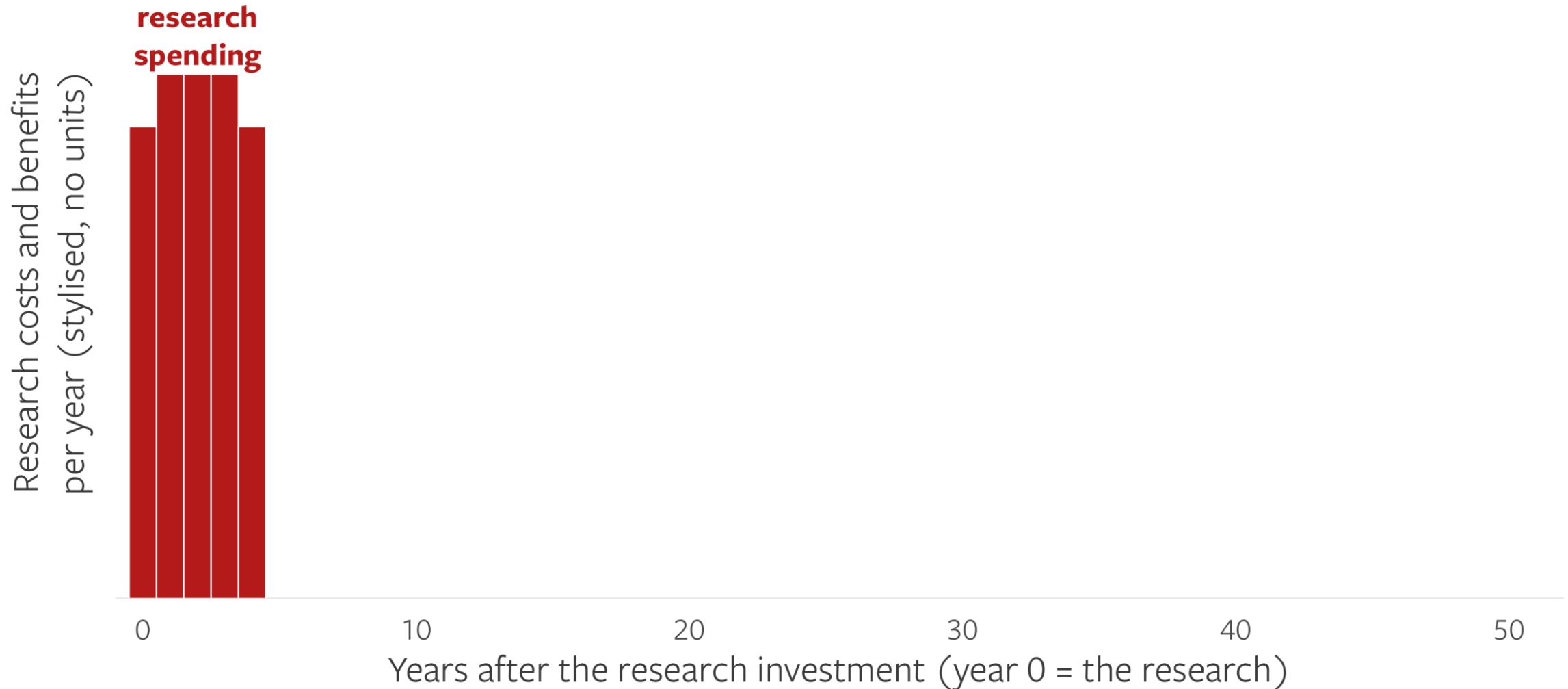
# Public ag R&D: the United States versus others

**2015: China \$10.1B vs. US \$4.9B — about 2 to 1**



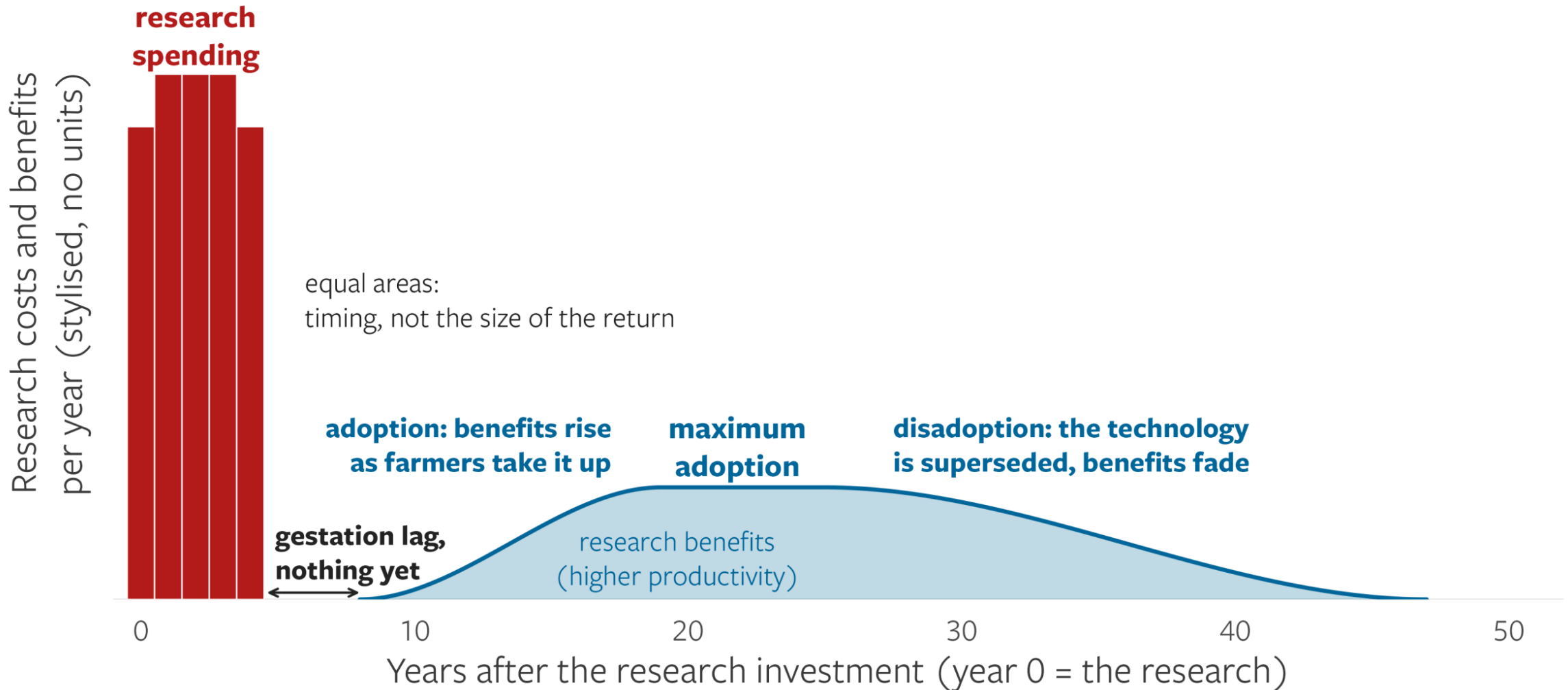
Source: USDA ERS chart of note (2022) based on ERS, ASTI/IFPRI and World Bank data; values digitized from the published chart.

# Research is an investment: the money goes in first...



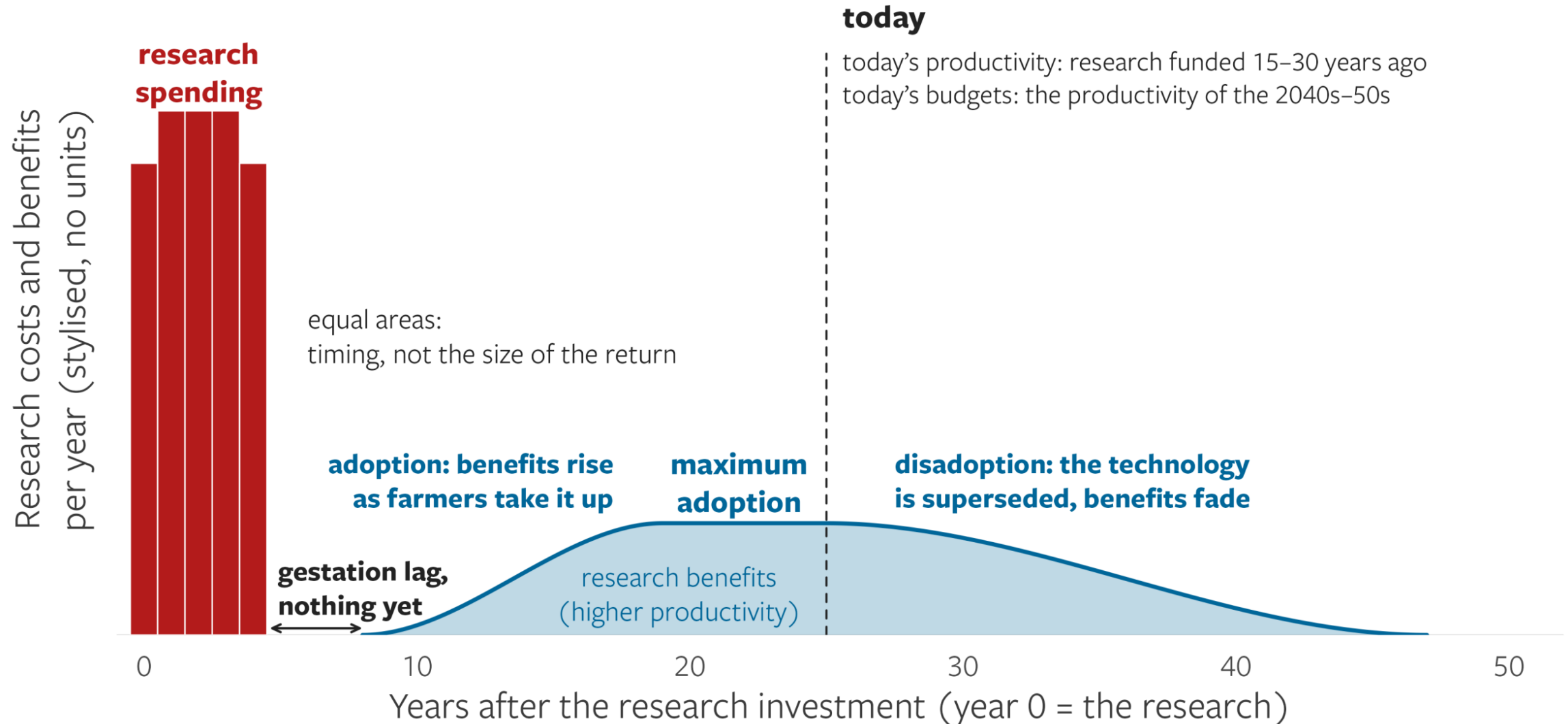
Stylised after Alston, Norton & Pardey (1995) and Alston et al. (2010); shape illustrative, not estimated; the two areas are drawn equal.

## ...and the benefits arrive over decades



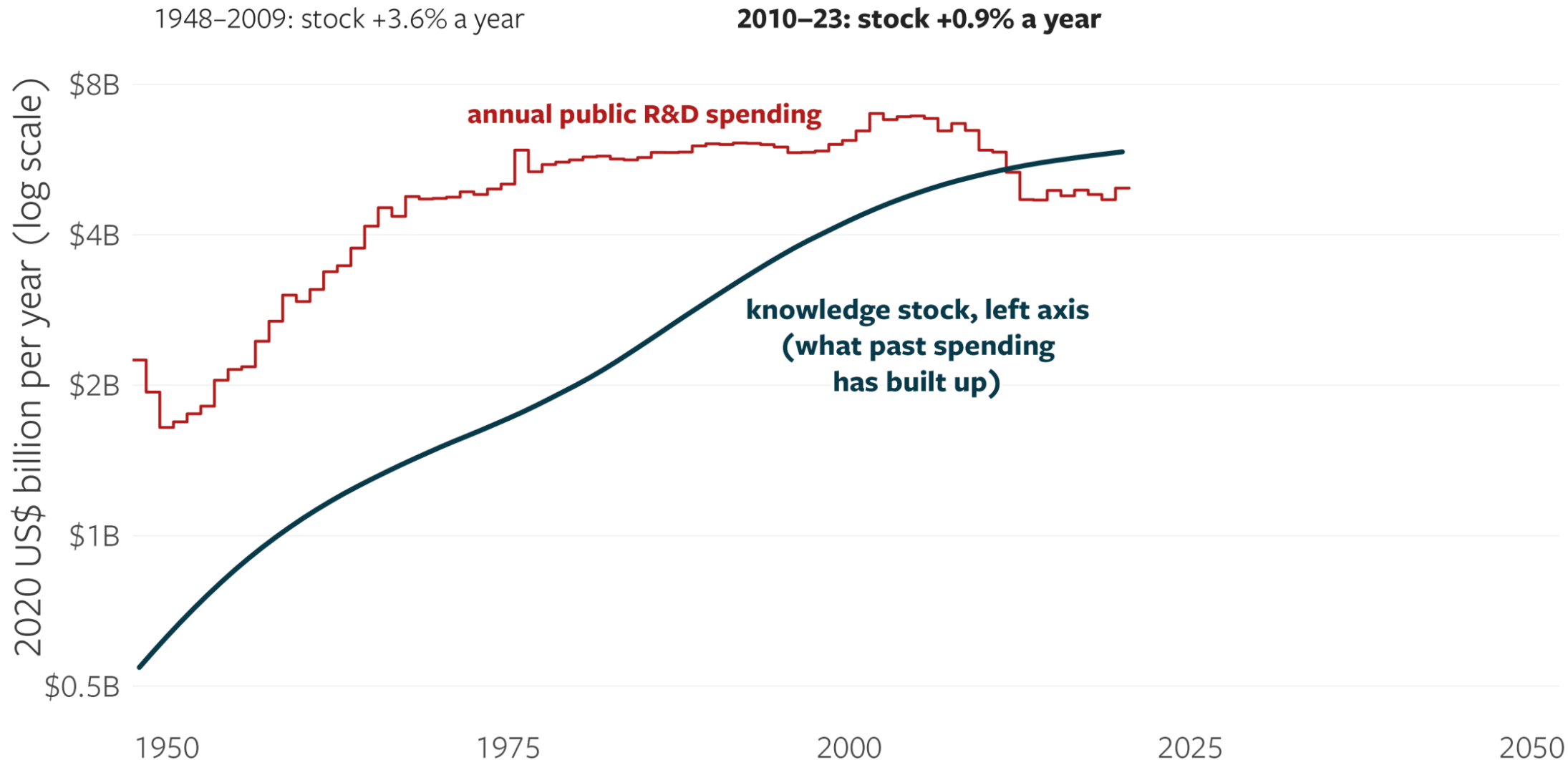
Stylised after Alston, Norton & Pardey (1995) and Alston et al. (2010); shape illustrative, not estimated; the two areas are drawn equal.

# Today's productivity reflects investments decades ago



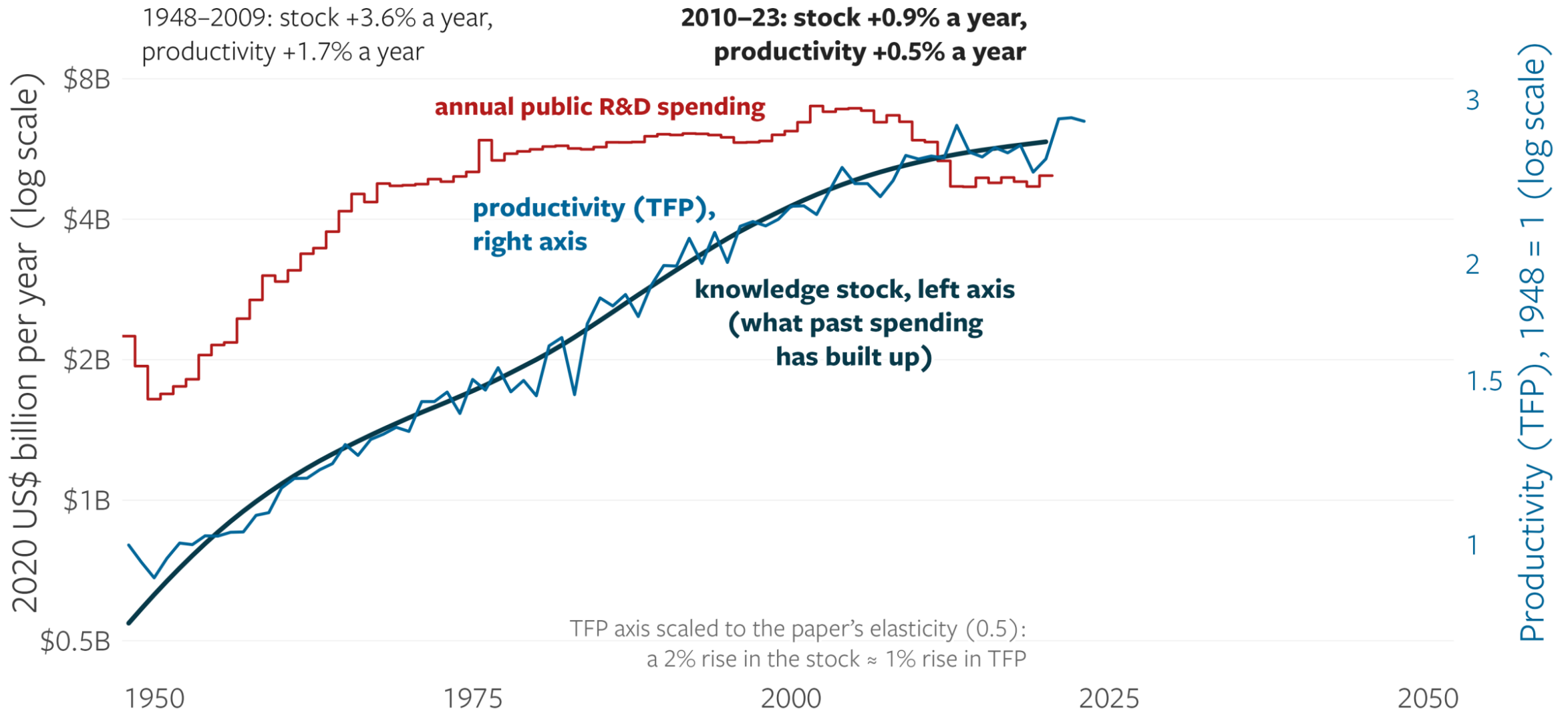
Stylised after Alston, Norton & Pardey (1995) and Alston et al. (2010); shape illustrative, not estimated; the two areas are drawn equal.

# Past spending adds up to a knowledge stock — which has stopped growing



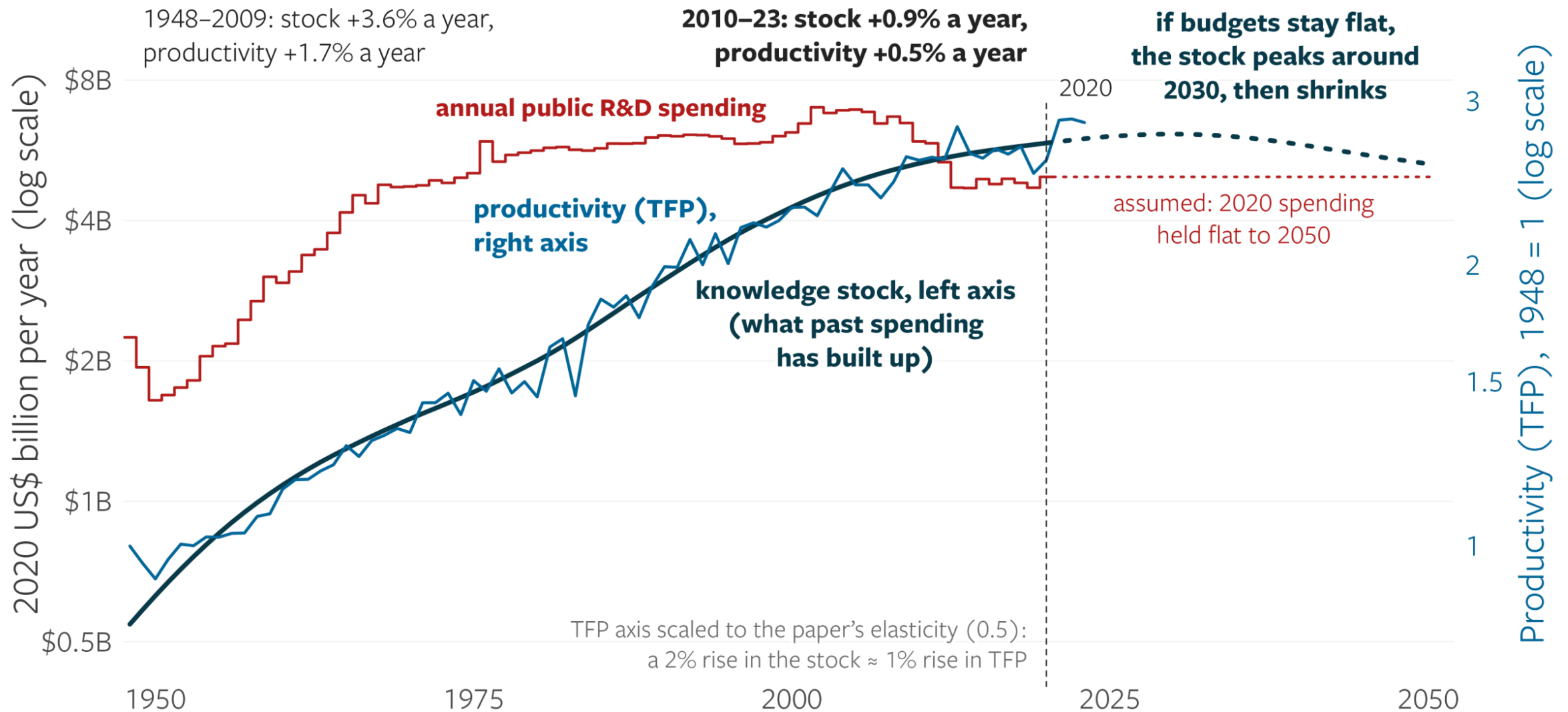
Source: USDA ERS spending and TFP; stock = 50-year weighted average of past spending with the baseline lag weights; TFP axis at elasticity 0.5 (Ortiz-Bobea et al. 2025).

# Productivity tracks the knowledge stock

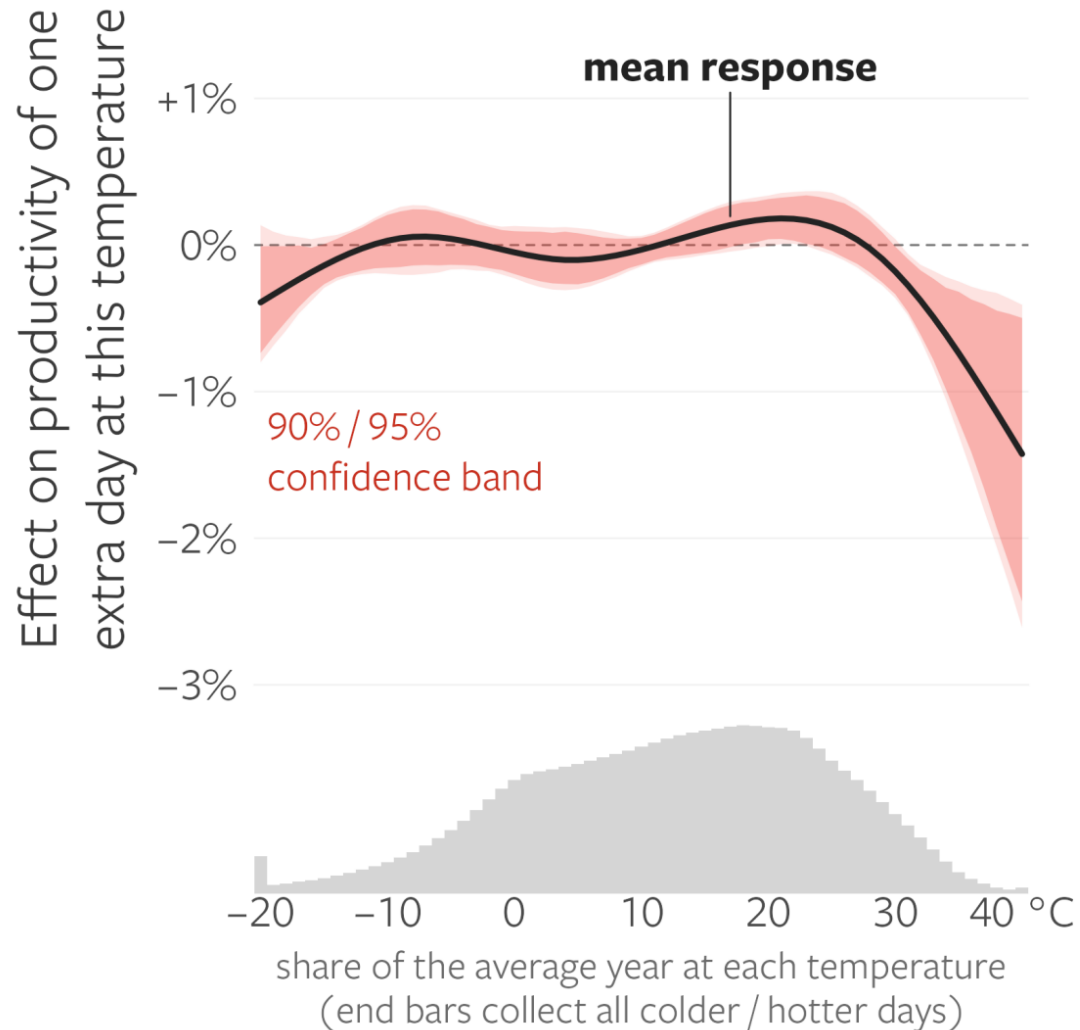


Source: USDA ERS spending and TFP; stock = 50-year weighted average of past spending with the baseline lag weights; TFP axis at elasticity 0.5 (Ortiz-Bobea et al. 2025).

# Flat budgets: the stock peaks around 2030, then shrinks

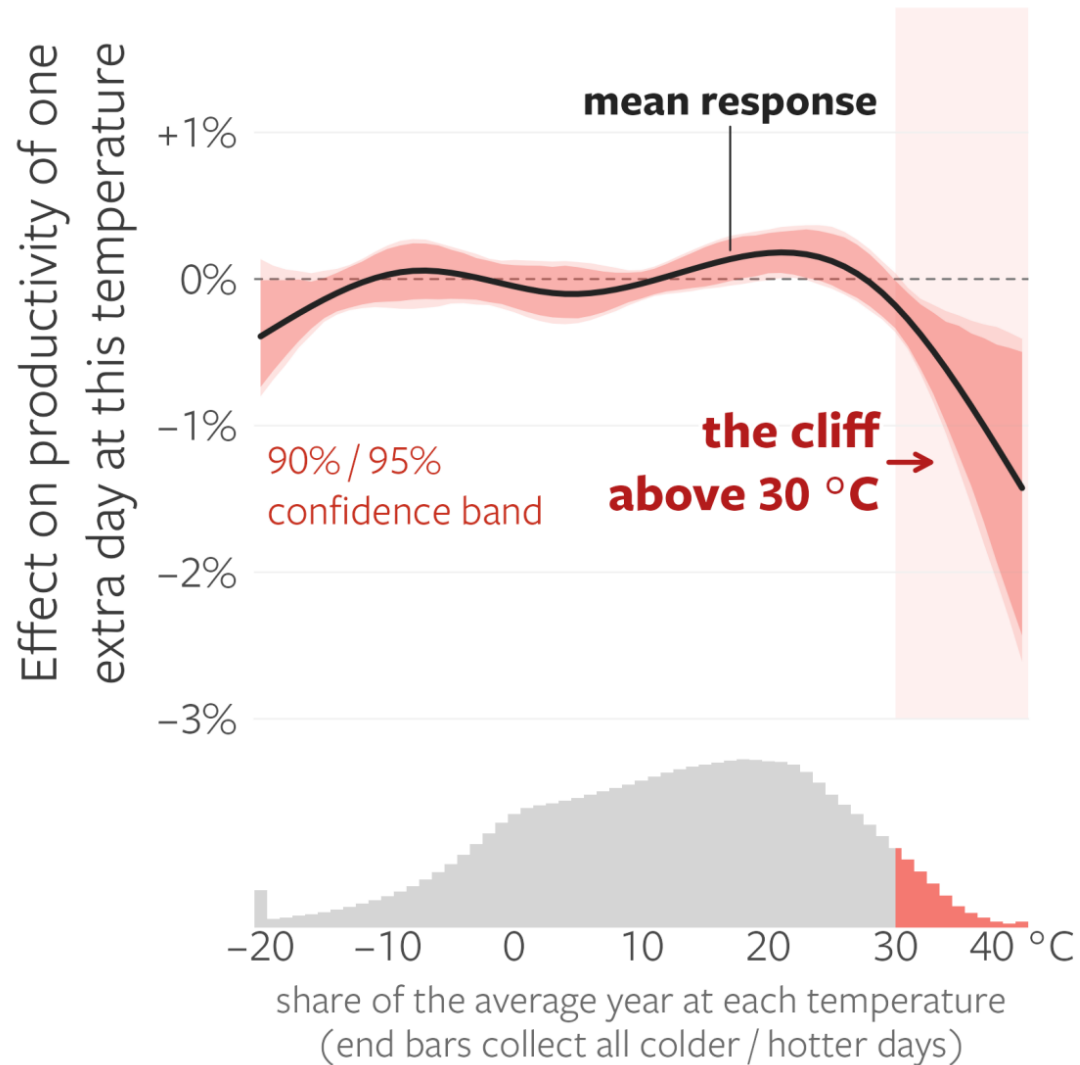


# Increasingly, it is climate: heat hurts productivity



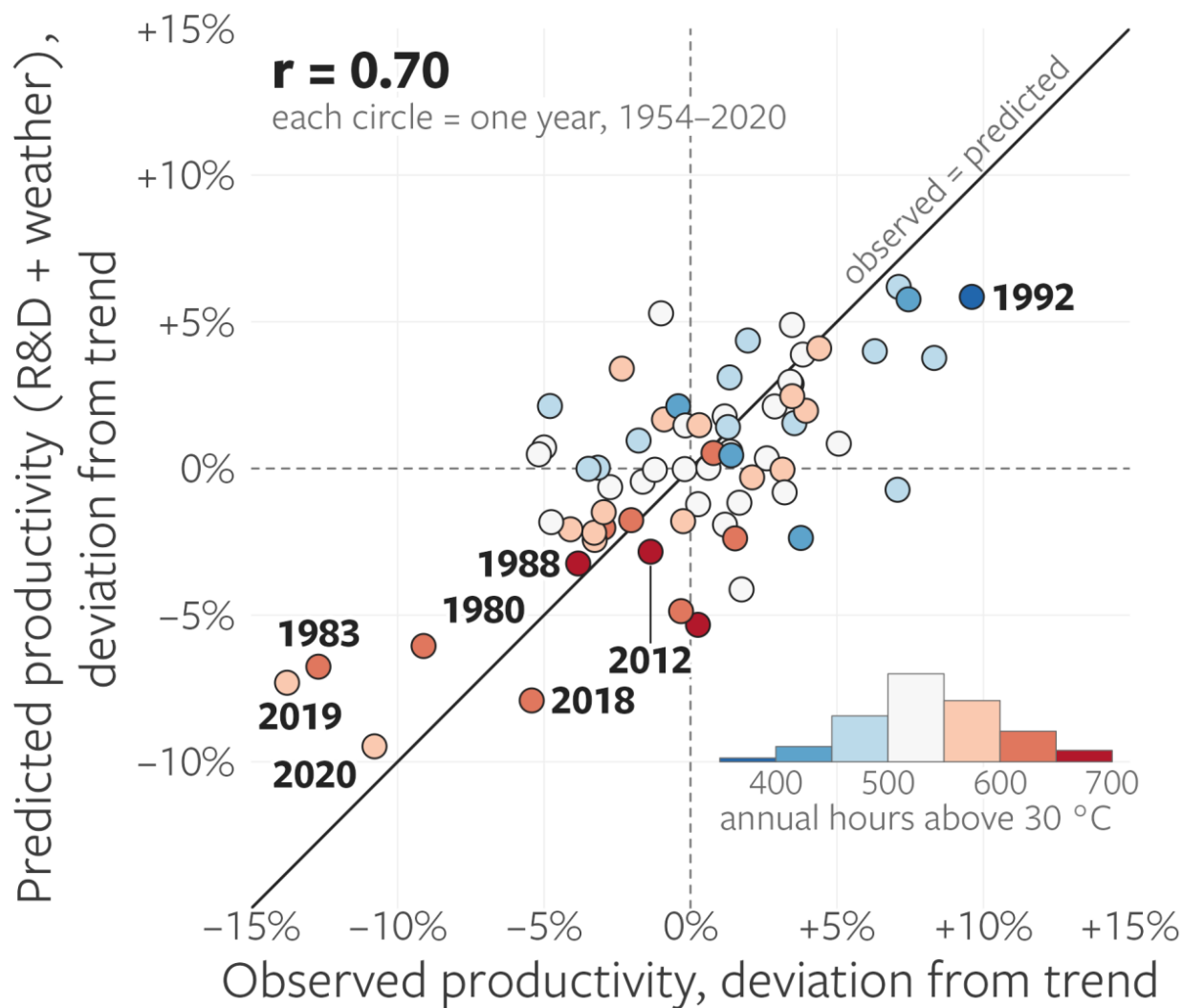
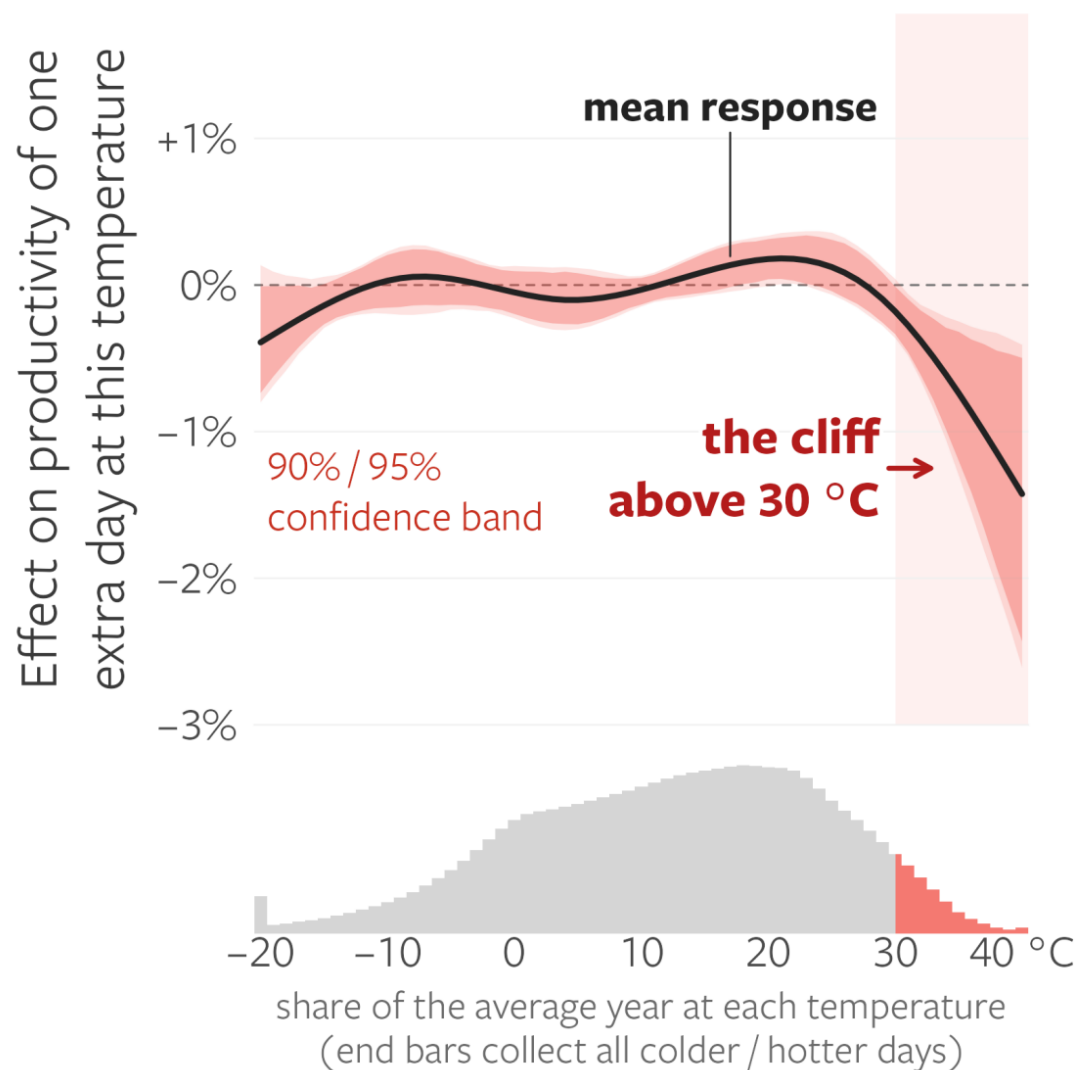
Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 2A-B.

# Above 30 °C, productivity falls off a cliff



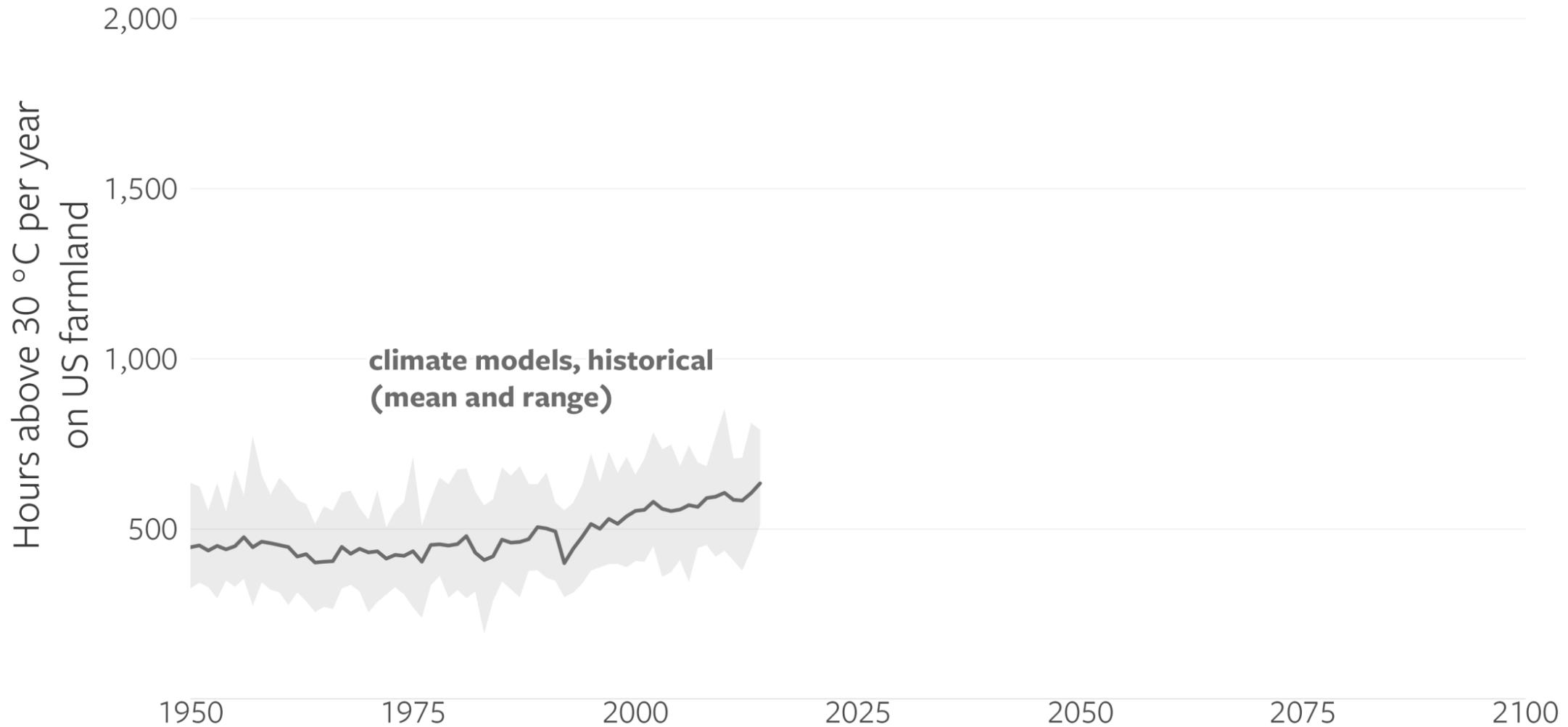
Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 2A-B.

# And in the data: the hot years are the bad years



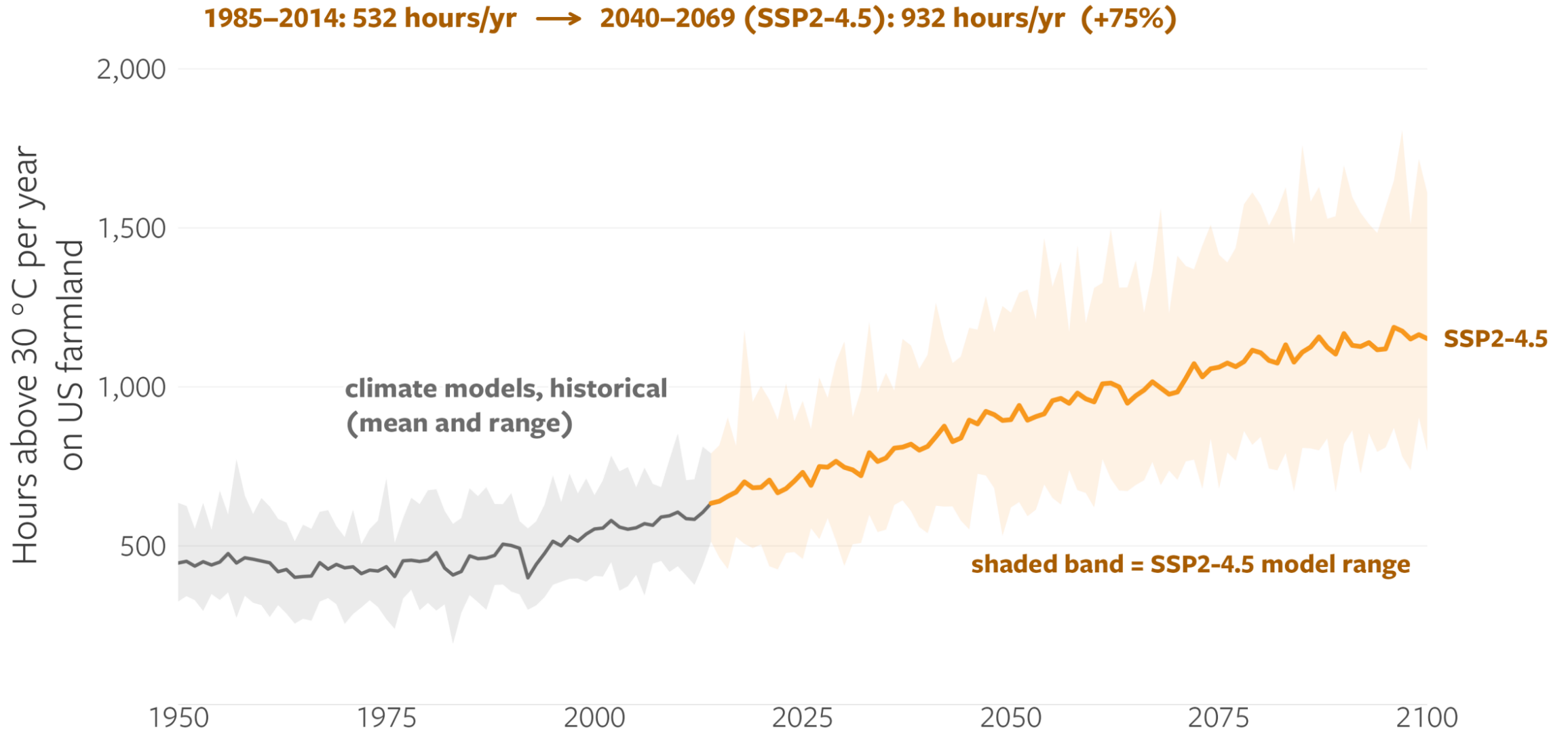
Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 2A–B.

# Hours above 30 °C over US farmland: the climate models' history



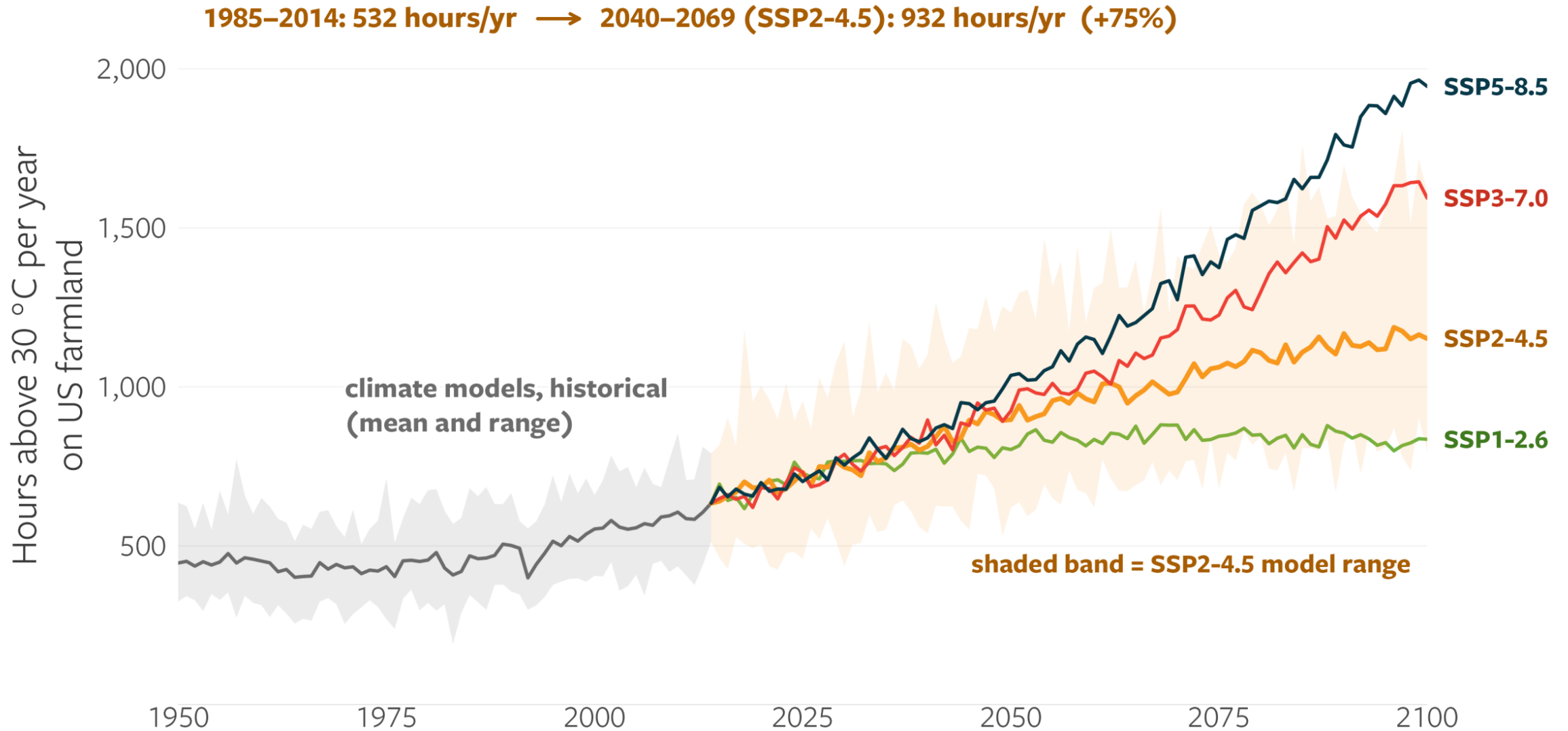
Source: Ortiz-Bobea et al. (PNAS 2025); 25 CMIP6 models (CIL downscaling), ensemble mean and range.

# Climate is pushing crops toward the cliff



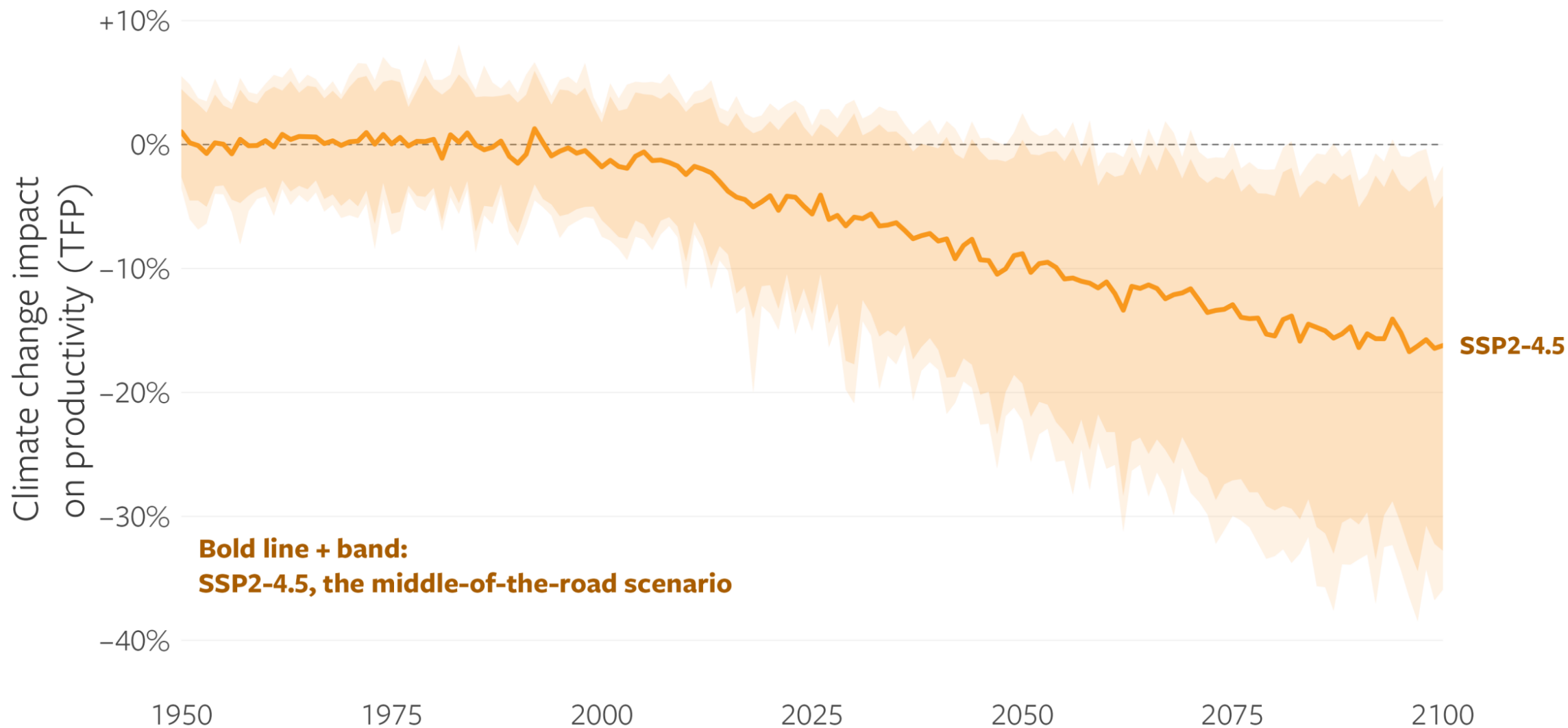
Source: Ortiz-Bobea et al. (PNAS 2025); 25 CMIP6 models (CIL downscaling), ensemble mean and range.

# Under every scenario



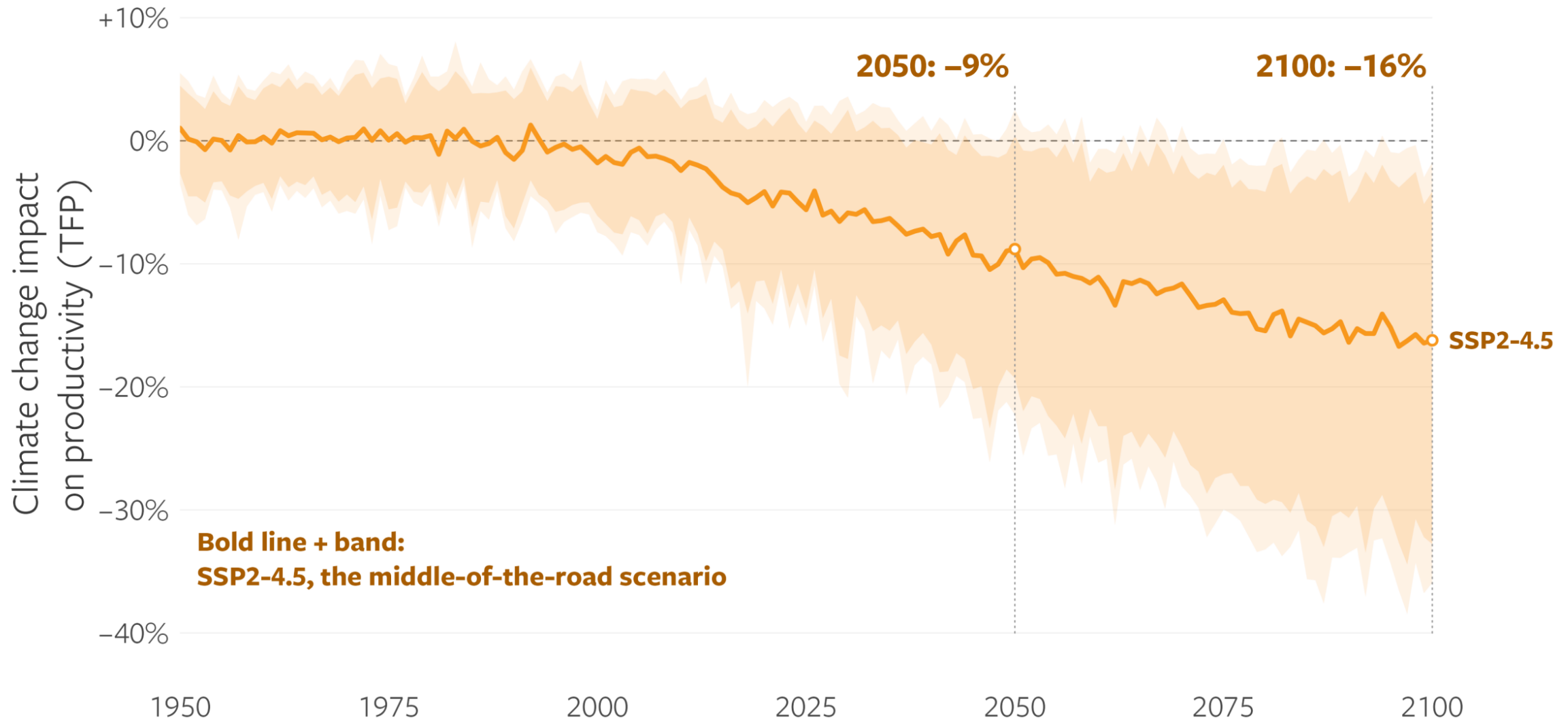
Source: Ortiz-Bobea et al. (PNAS 2025); 25 CMIP6 models (CIL downscaling), ensemble mean and range.

# What that does to productivity



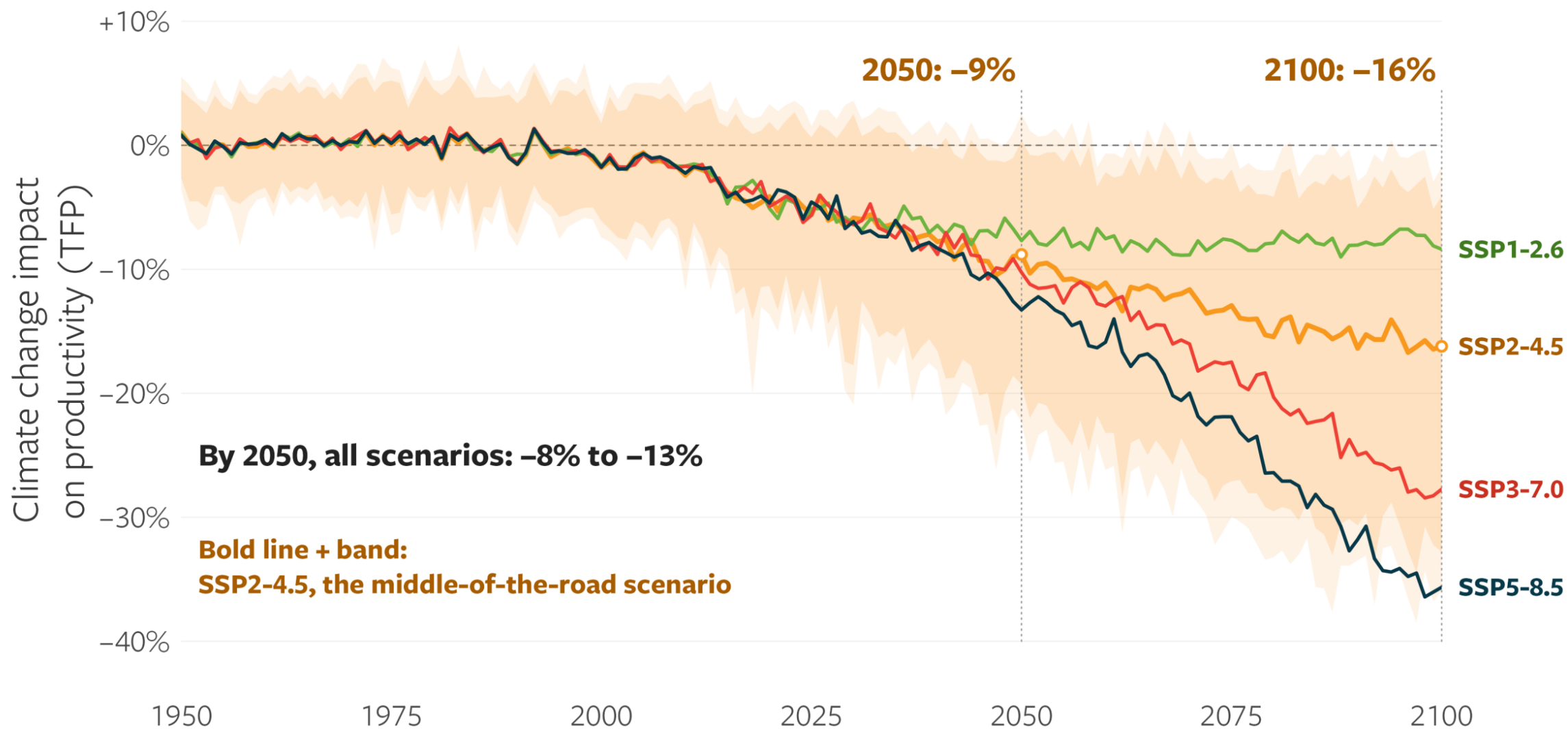
Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 3: CMIP6 ensemble with bootstrap uncertainty (90% and 95% bands).

# About 9% lower by 2050



Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 3: CMIP6 ensemble with bootstrap uncertainty (90% and 95% bands).

# 8–13% lower by 2050 — under every scenario



Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 3: CMIP6 ensemble with bootstrap uncertainty (90% and 95% bands).

## Message 2 — recap

- 1 Productivity growth is slowing.
- 2 **The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.

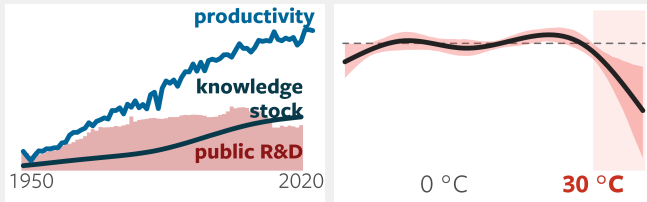
## Message 3

- 1 Productivity growth is slowing.
- 2 The main driver is slowing R&D — but increasingly it's climate.
- 3 **Maintaining productivity growth requires a historical pivot.**

# What we did in the PNAS 2025 paper, in three steps

## STEP 1

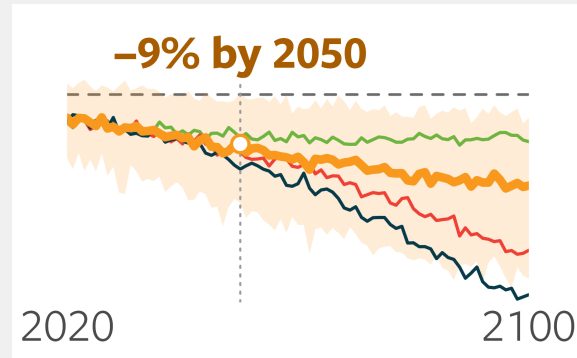
### History: R&D, heat and productivity



Seventy years of data: what each research dollar bought, with what lags — and how heat cuts productivity, with a cliff above 30 °C

## STEP 2

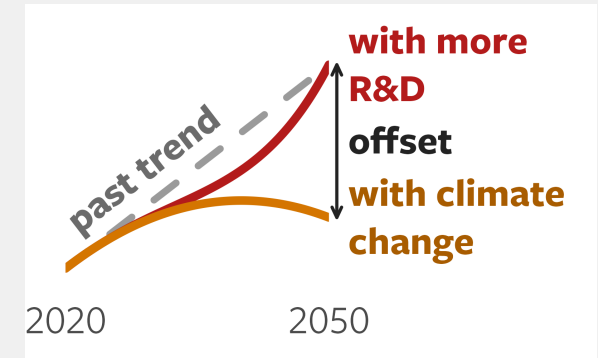
### The climate slowdown



Run 25 climate models and 4 scenarios through that response: about 9% lower productivity by 2050

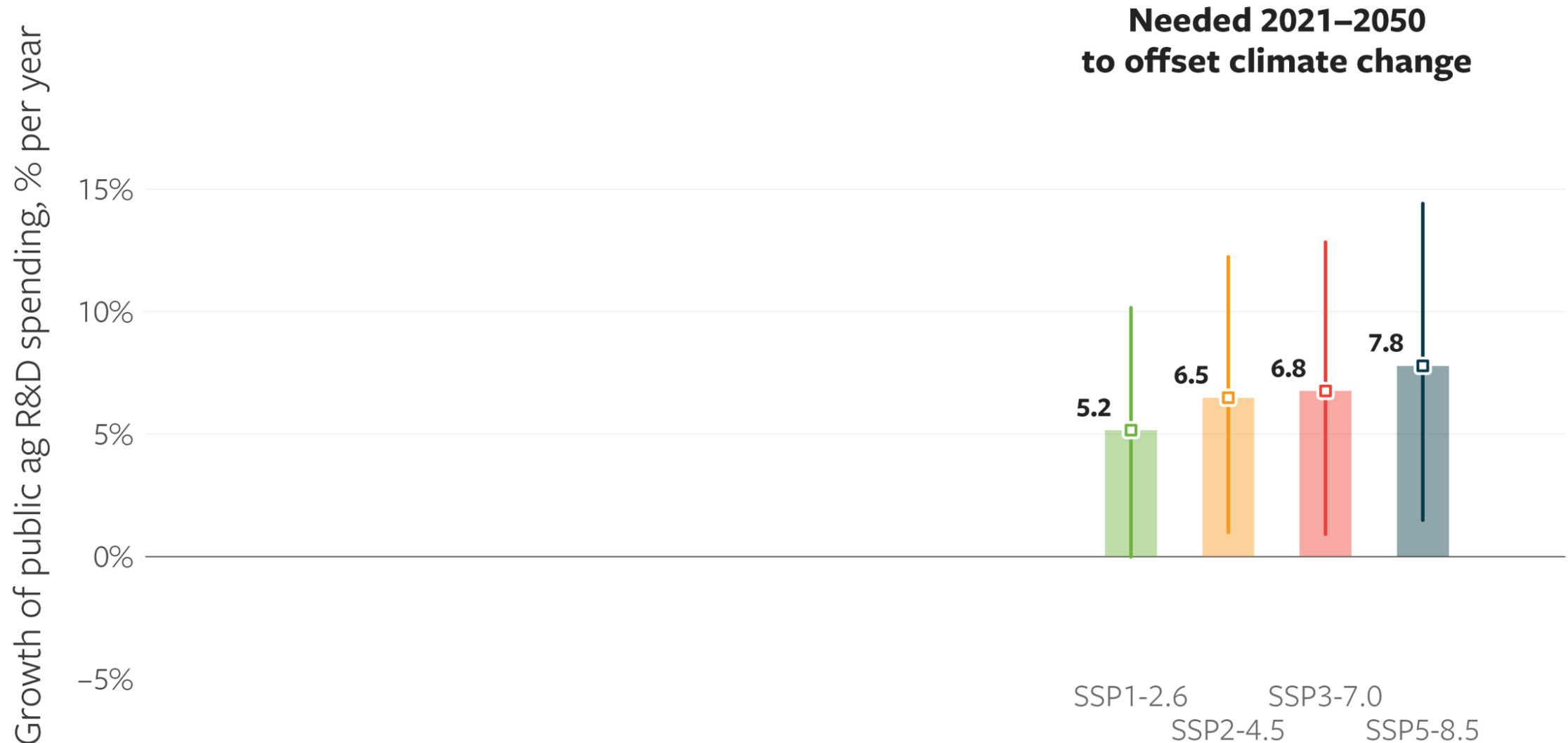
## STEP 3

### Solve for the R&D



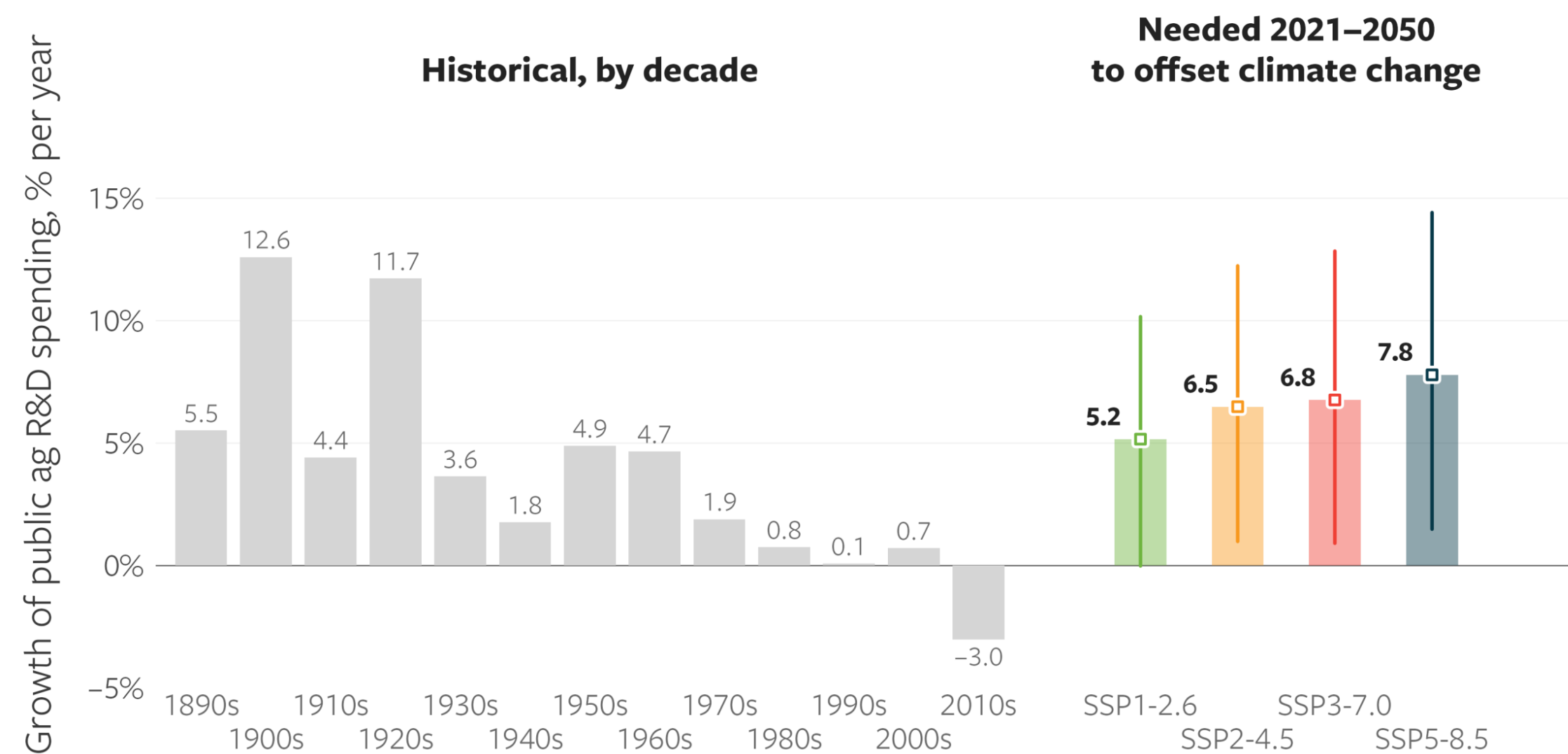
The research spending that keeps productivity on its past trend, to 2050

# Strategy 1: grow the budget gradually — offsetting climate by 2050 takes 5–8% a year



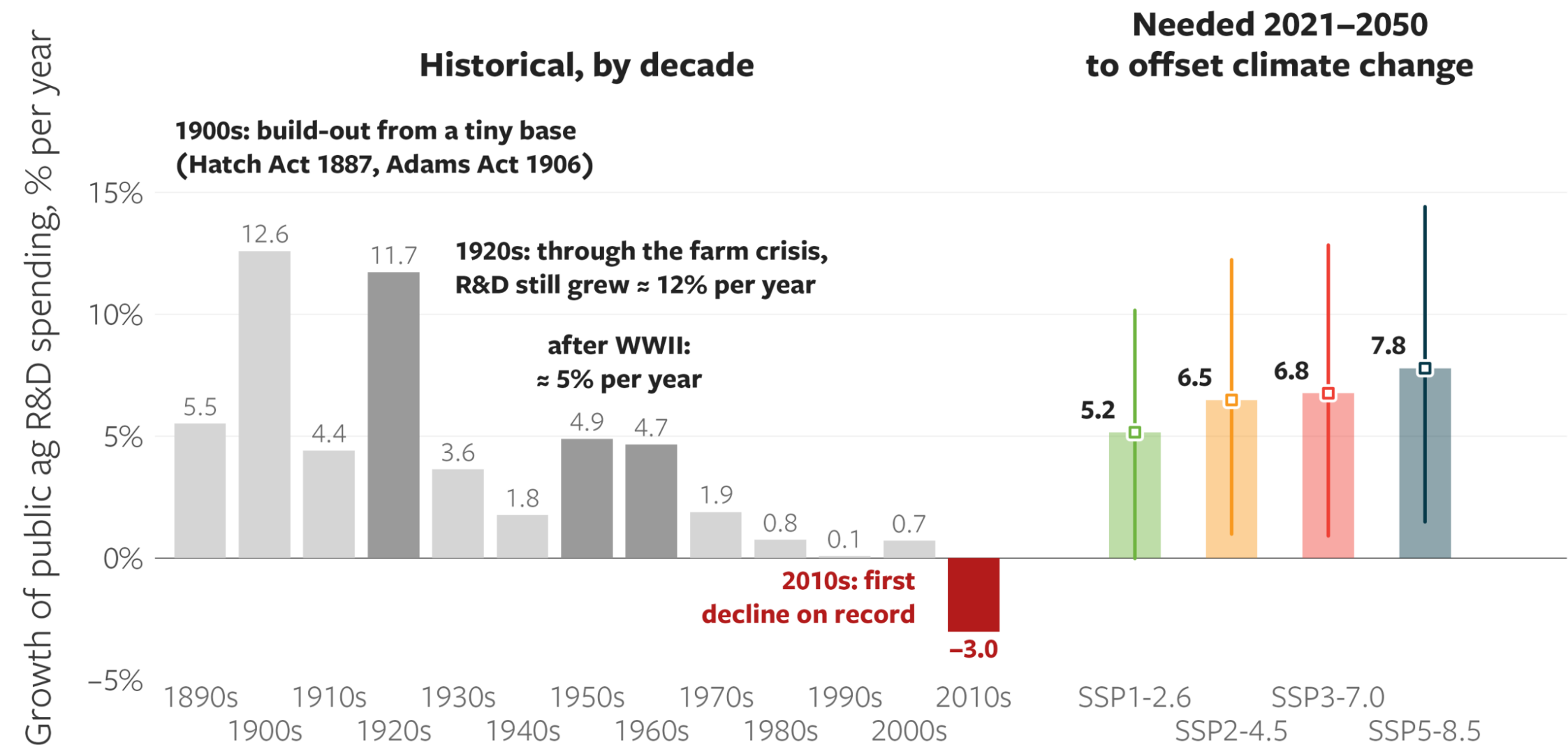
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4B and Table S12 (means and 95% CI).

# Has the budget ever grown that fast? Public ag R&D growth, by decade

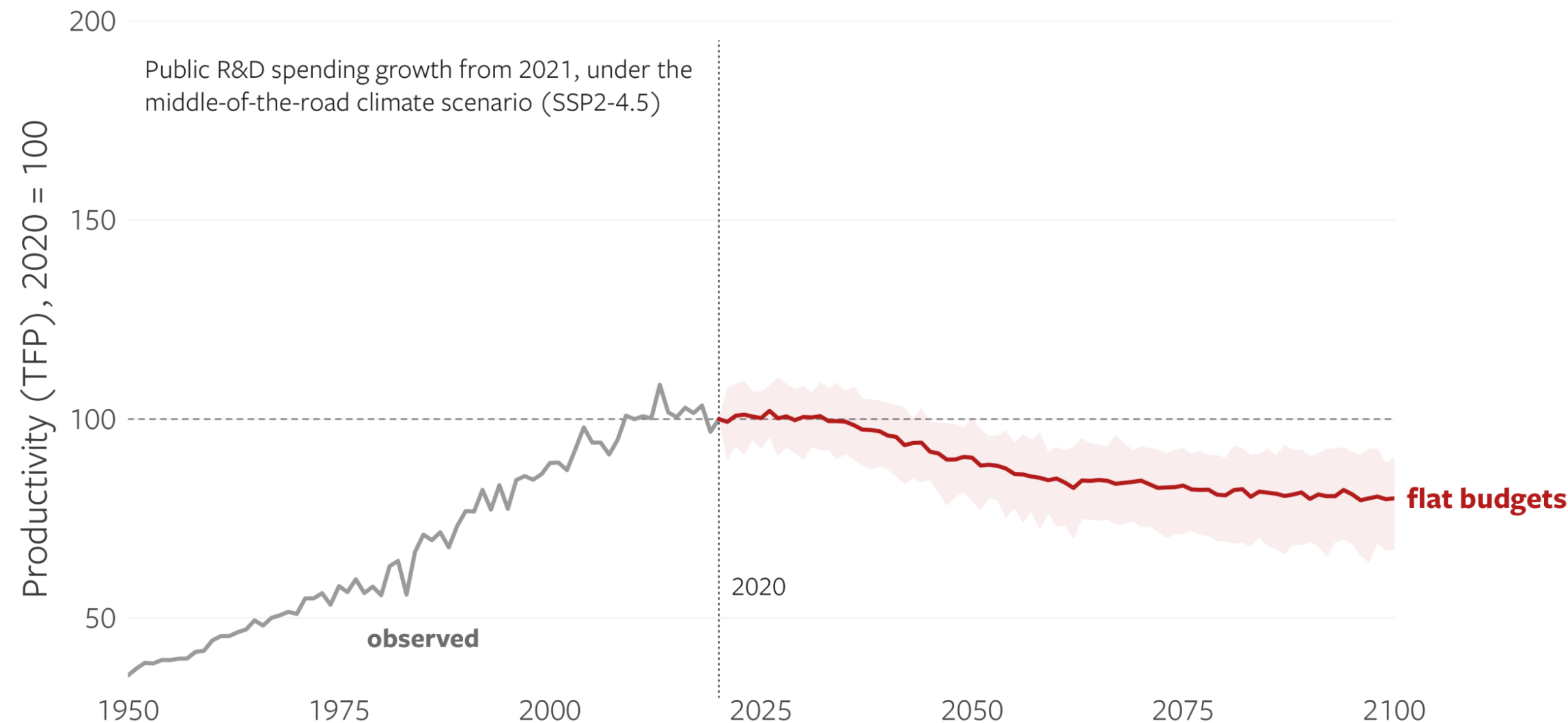


Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4B and Table S12 (means and 95% CI).

# We have done this before. Twice — after each world war

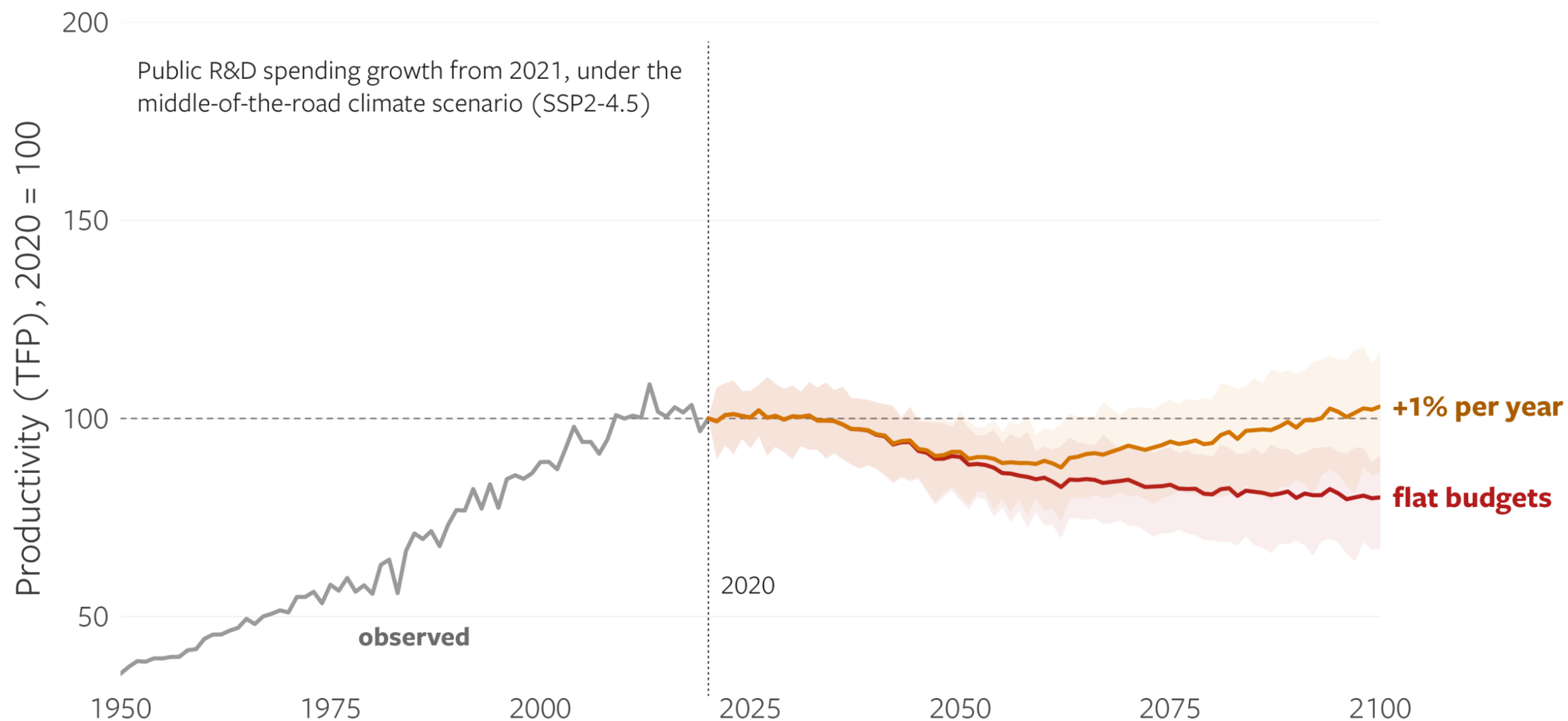


# Gradual R&D growth points to a delayed recovery



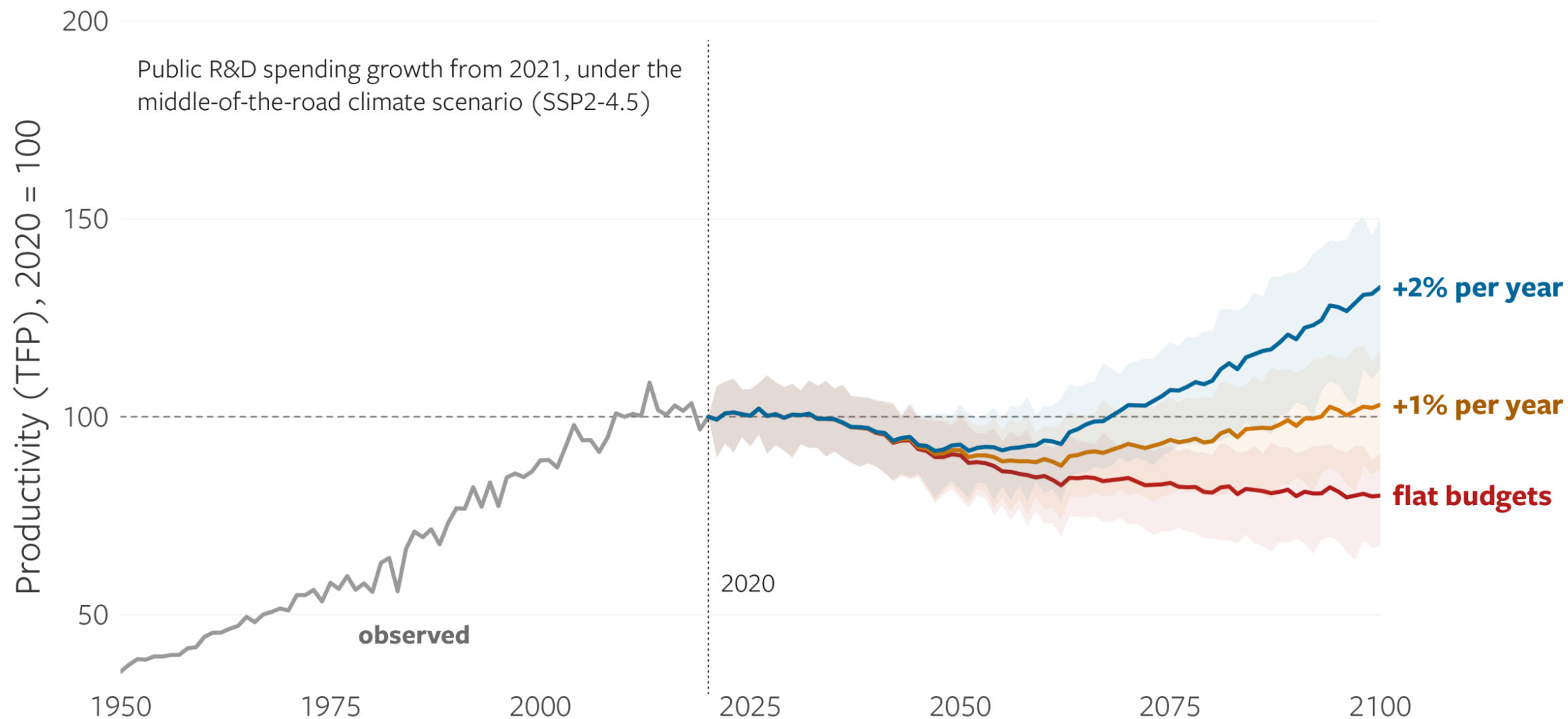
Source: Ortiz-Bobea et al. (PNAS 2025), projection following Fig. S26 (R&D lag structure, CMIP6 climate impacts); shaded = 90% range.

# Gradual R&D growth points to a delayed recovery



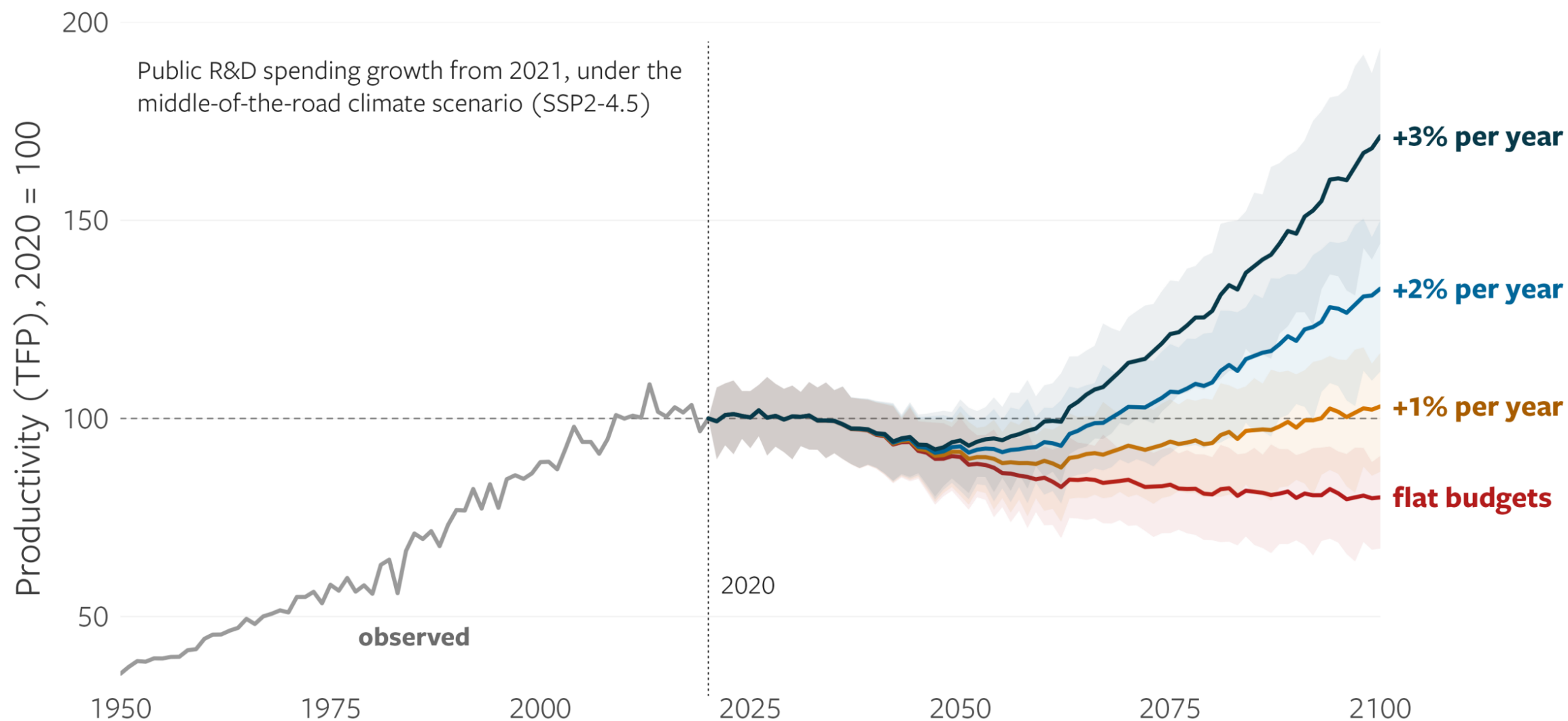
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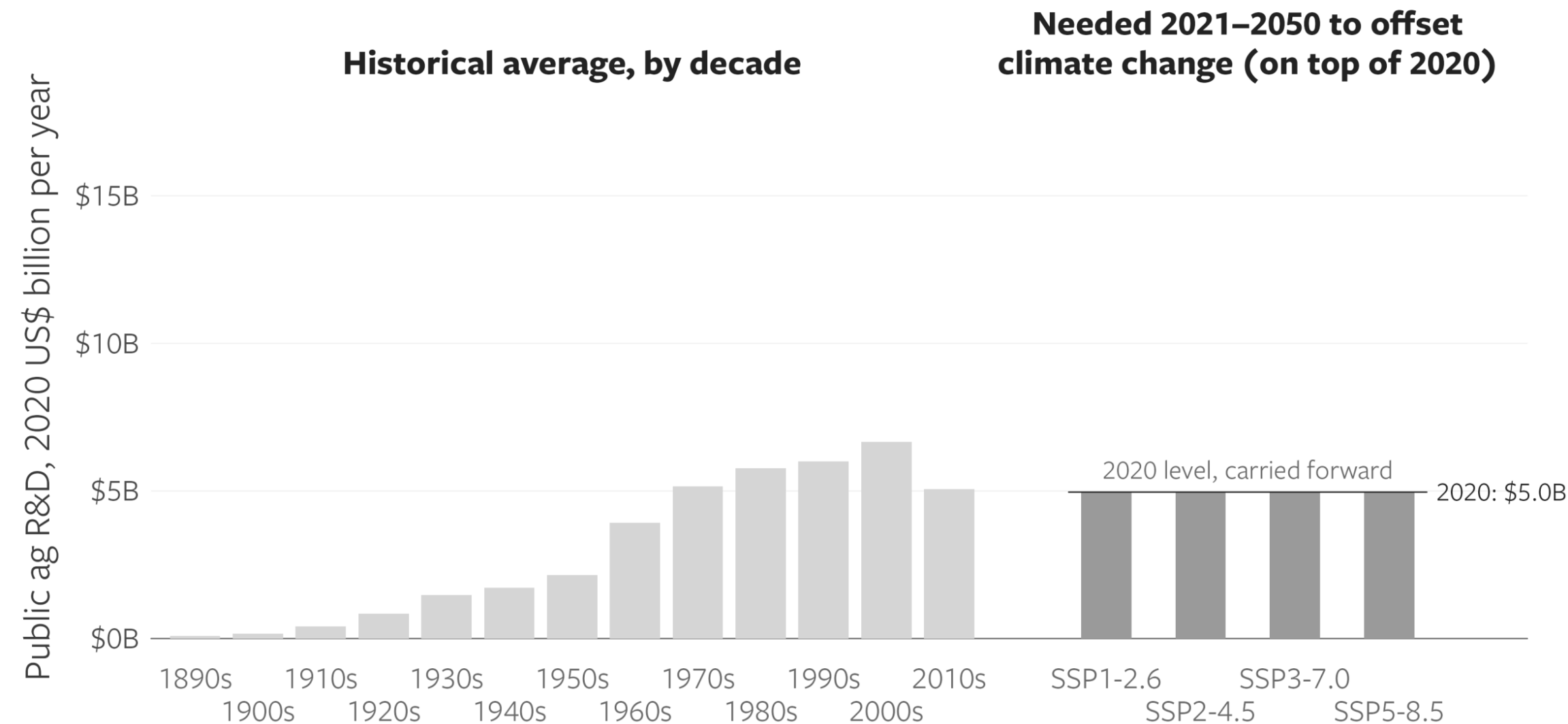
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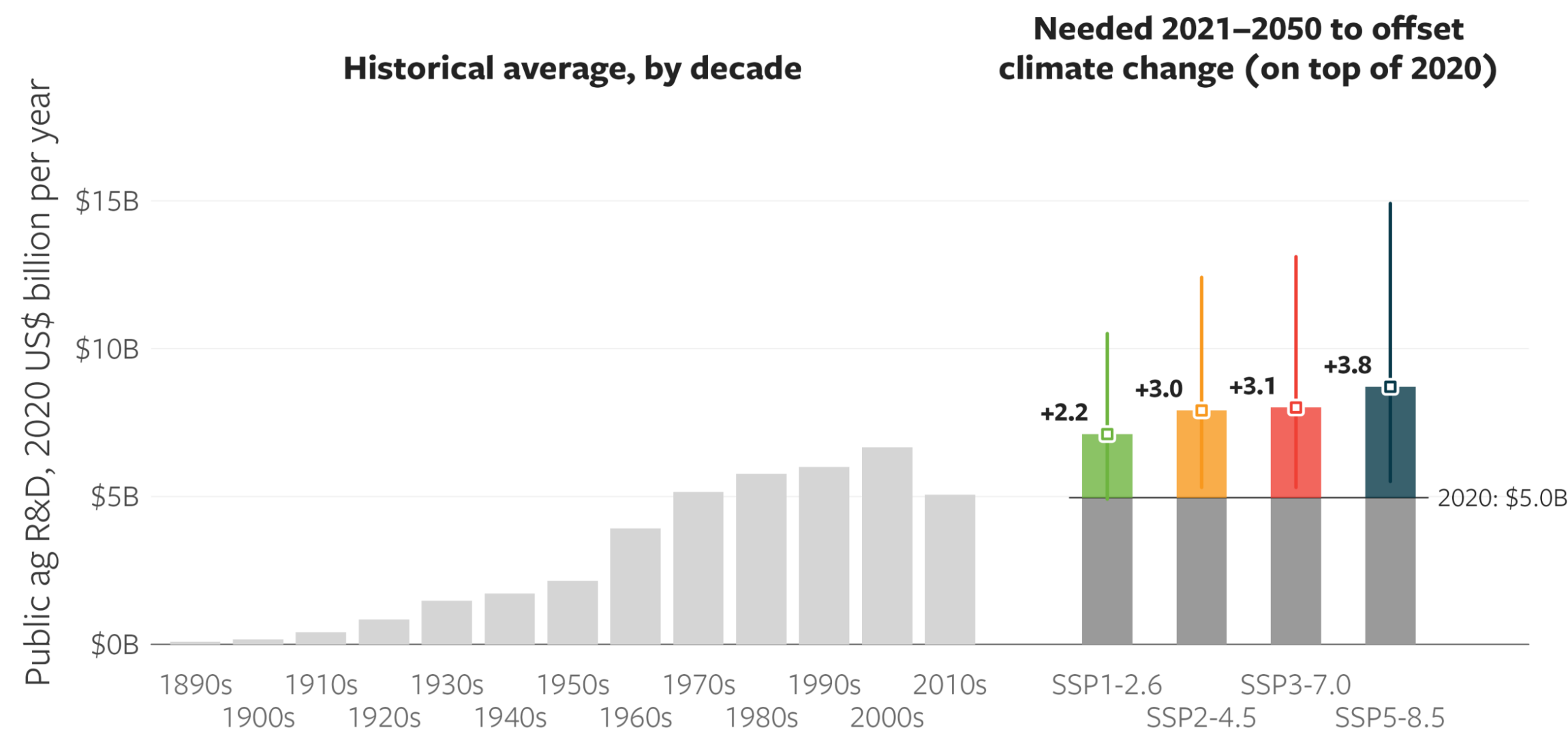
Source: Ortiz-Bobea et al. (PNAS 2025), projection following Fig. S26 (R&D lag structure, CMIP6 climate impacts); shaded = 90% range.

# Strategy 2: a fixed supplement on top of today's \$5B



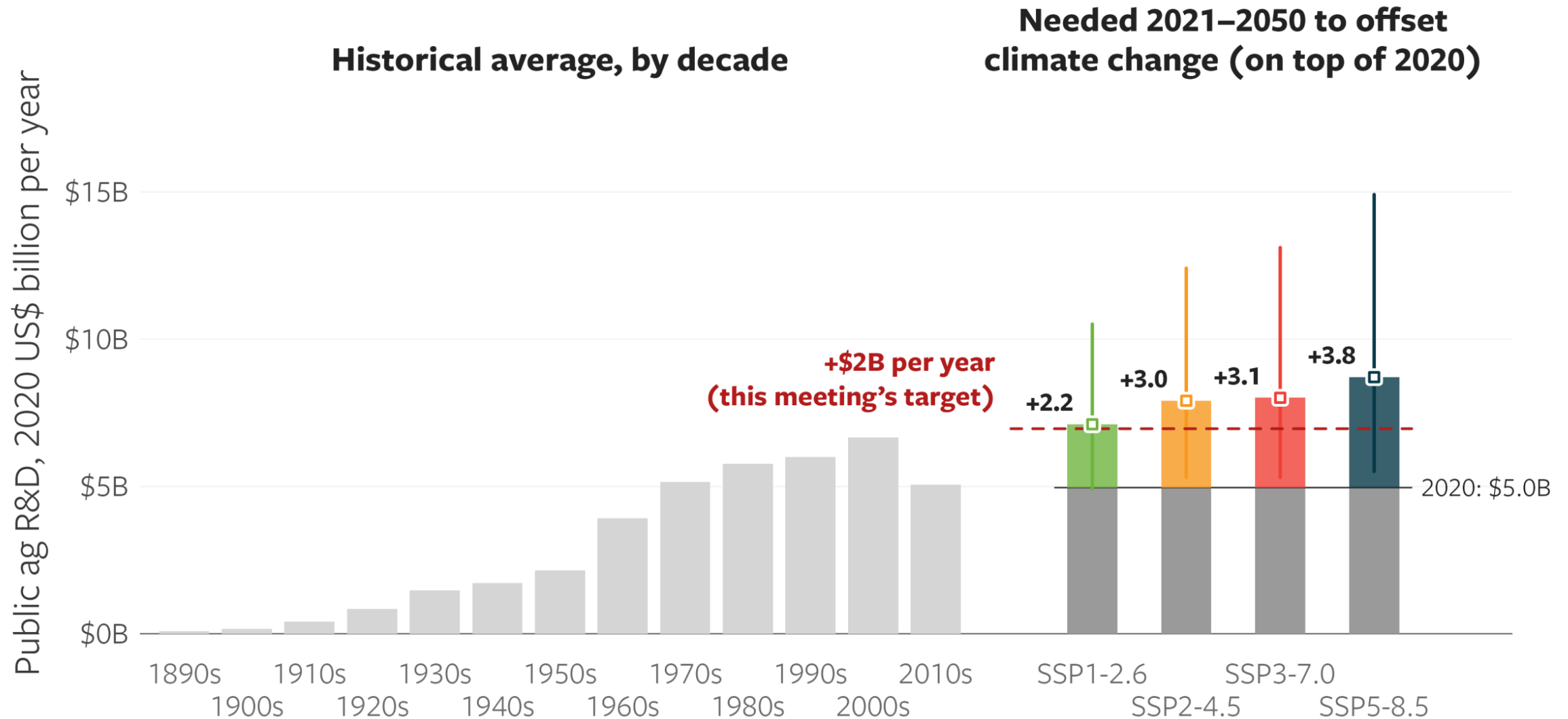
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14-S15 (fixed-supplement scenario, means and 95% CI).

# It takes \$2–4B more per year just to offset climate



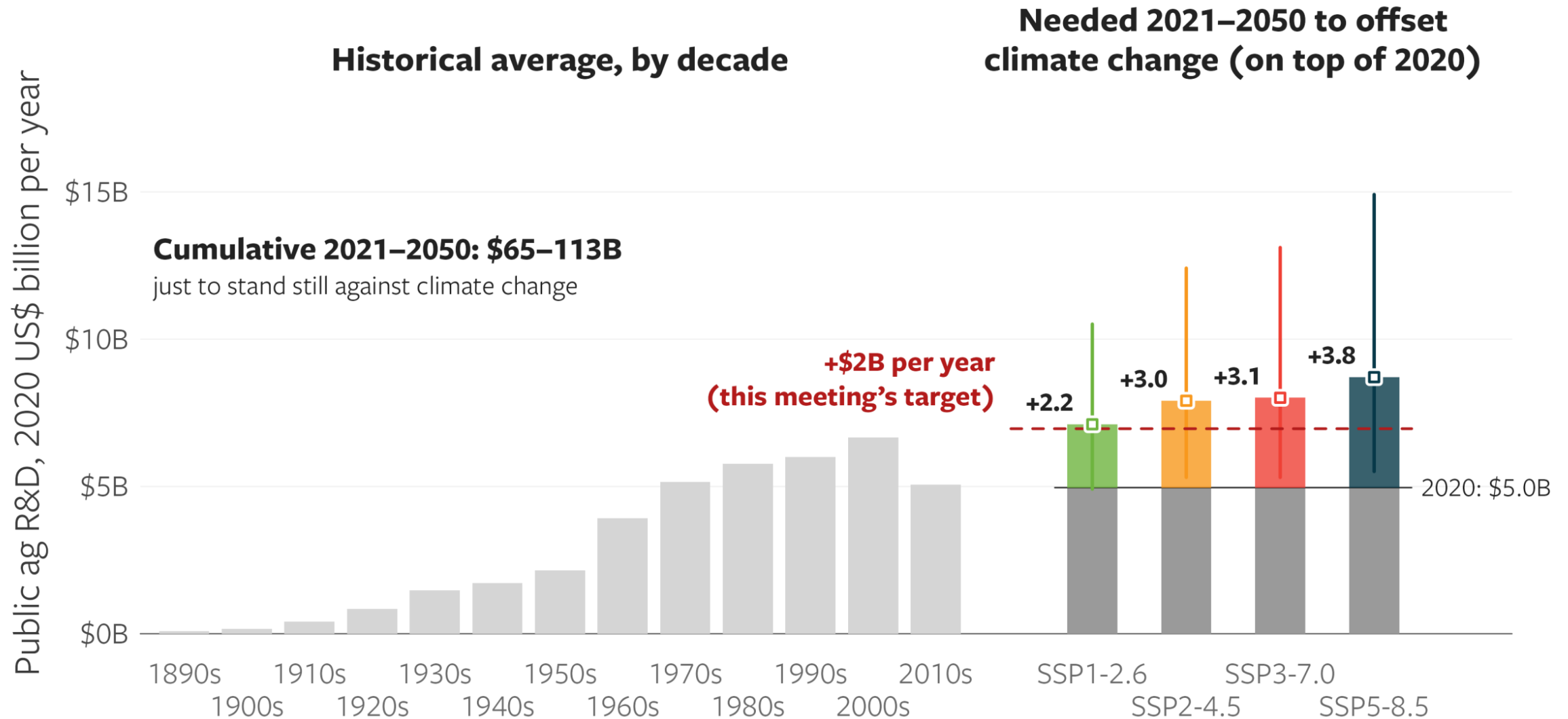
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14–S15 (fixed-supplement scenario, means and 95% CI).

# \$2B a year is the floor for standing still



Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14–S15 (fixed-supplement scenario, means and 95% CI).

# \$65–113B by 2050, just to stand still



Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14–S15 (fixed-supplement scenario, means and 95% CI).

# Limitations

## **No targeted adaptation research**

We price the historical mix of research; adaptation-focused R&D could be cheaper — conservative.

## **Future returns like the past**

We assume research pays off as it did over 1950–2020 — could cut either way.

## **Public R&D only**

Private R&D is a complement, not a substitute; it is not in this number.

# Key messages of this talk

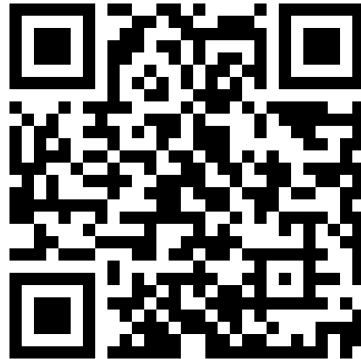
- 1 Productivity growth is slowing.**
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.**

# Thank you

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**the paper**



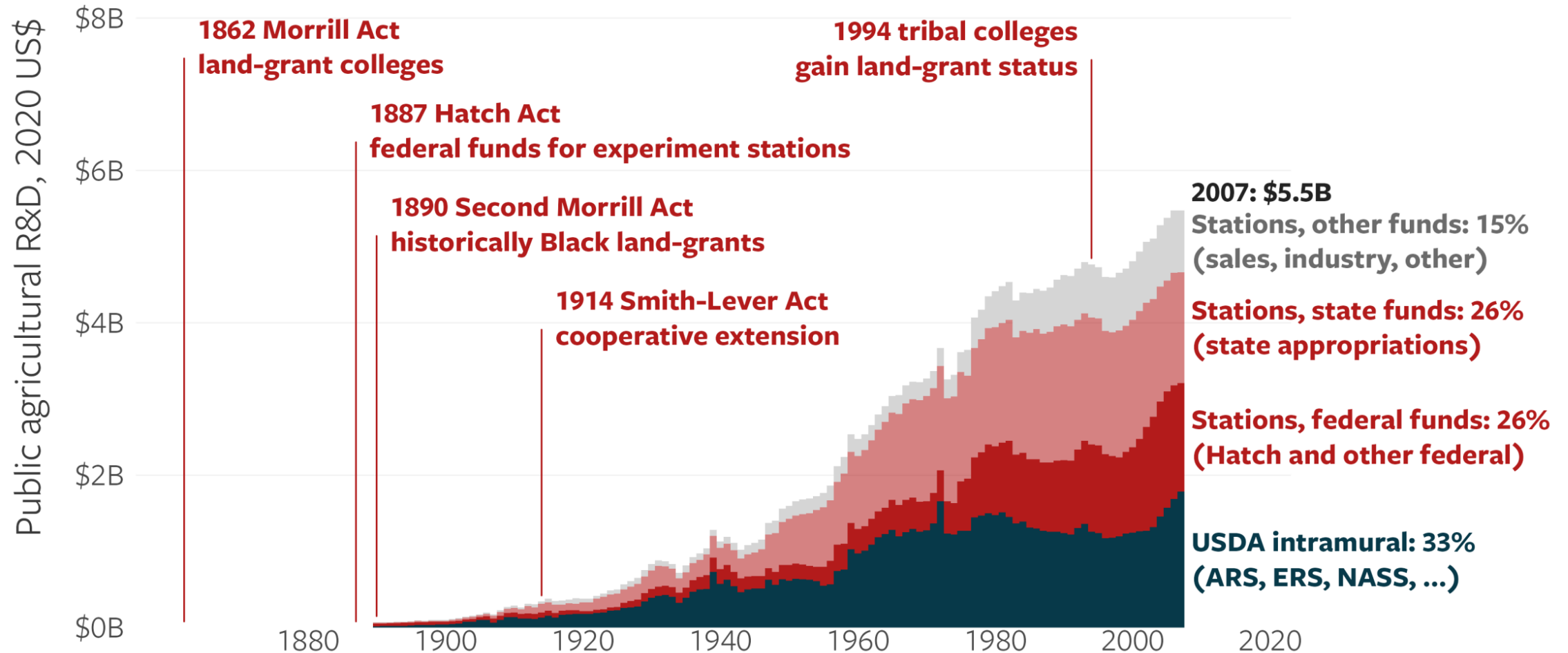
**these slides**

# Extra slides

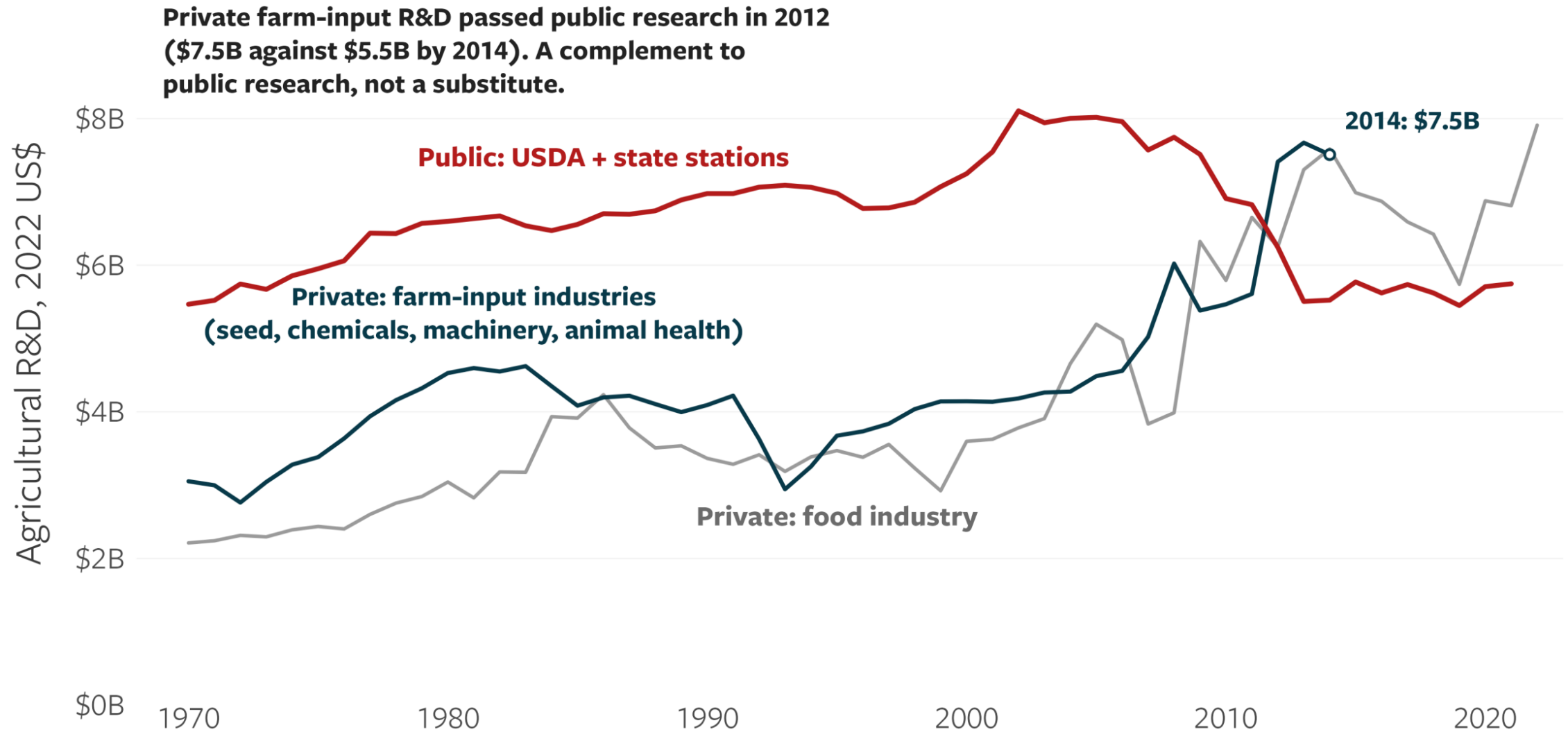
who pays for public research · public vs private R&D

# Who spends it, and who pays: USDA and the state stations, by source of funds

InSTePP accounts (Pardey et al.), 1890–2007: USDA intramural + experiment stations, by source of funds; excl. forestry and extension; levels differ slightly from the ERS series (deflator, vet schools).



# Private R&D has overtaken public research — and it is a complement, not a substitute



Source: USDA ERS, Agricultural and Food R&D Expenditures in the United States (1970–2022), constant 2022 US\$; private input-industry series ends in 2014.